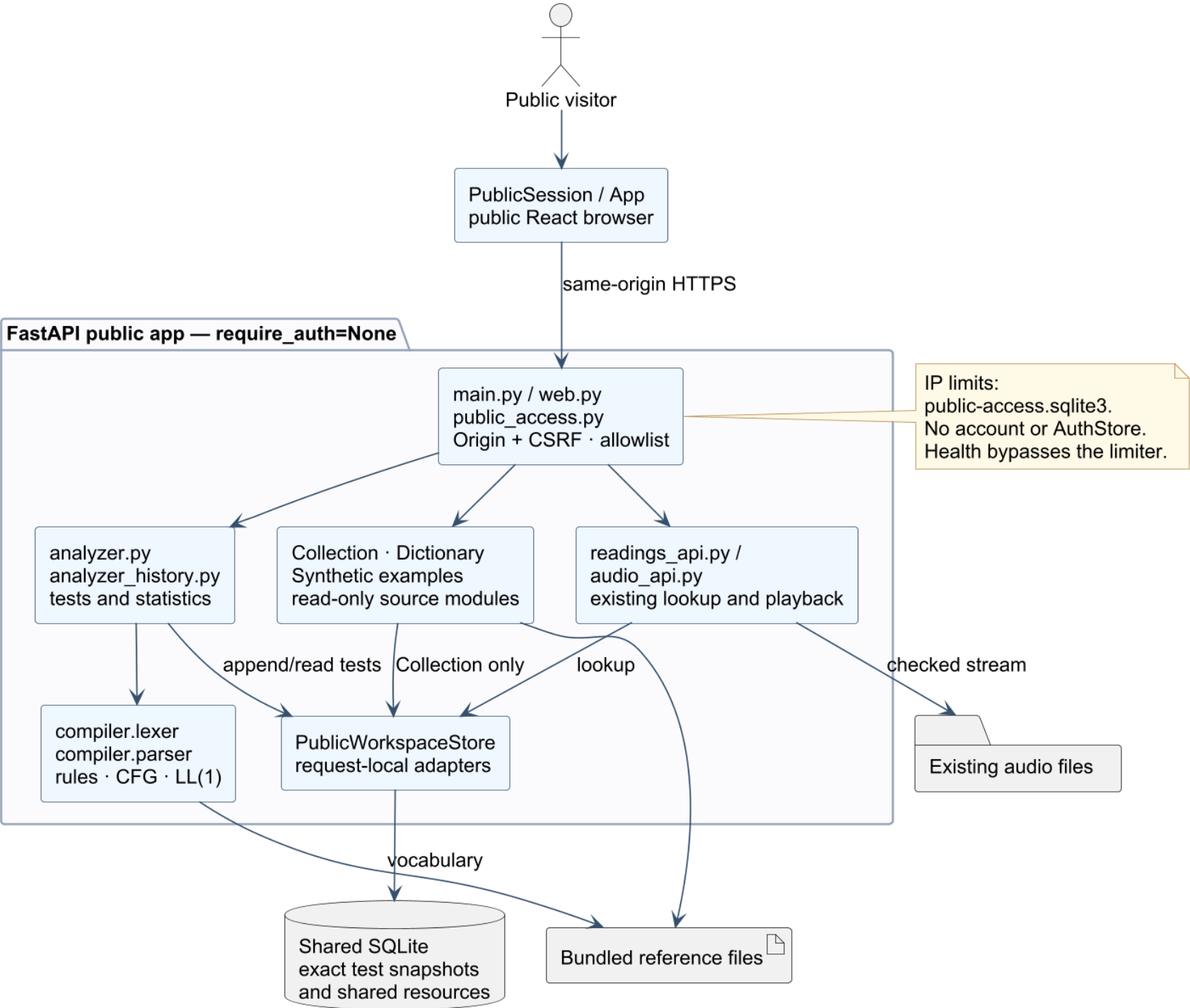
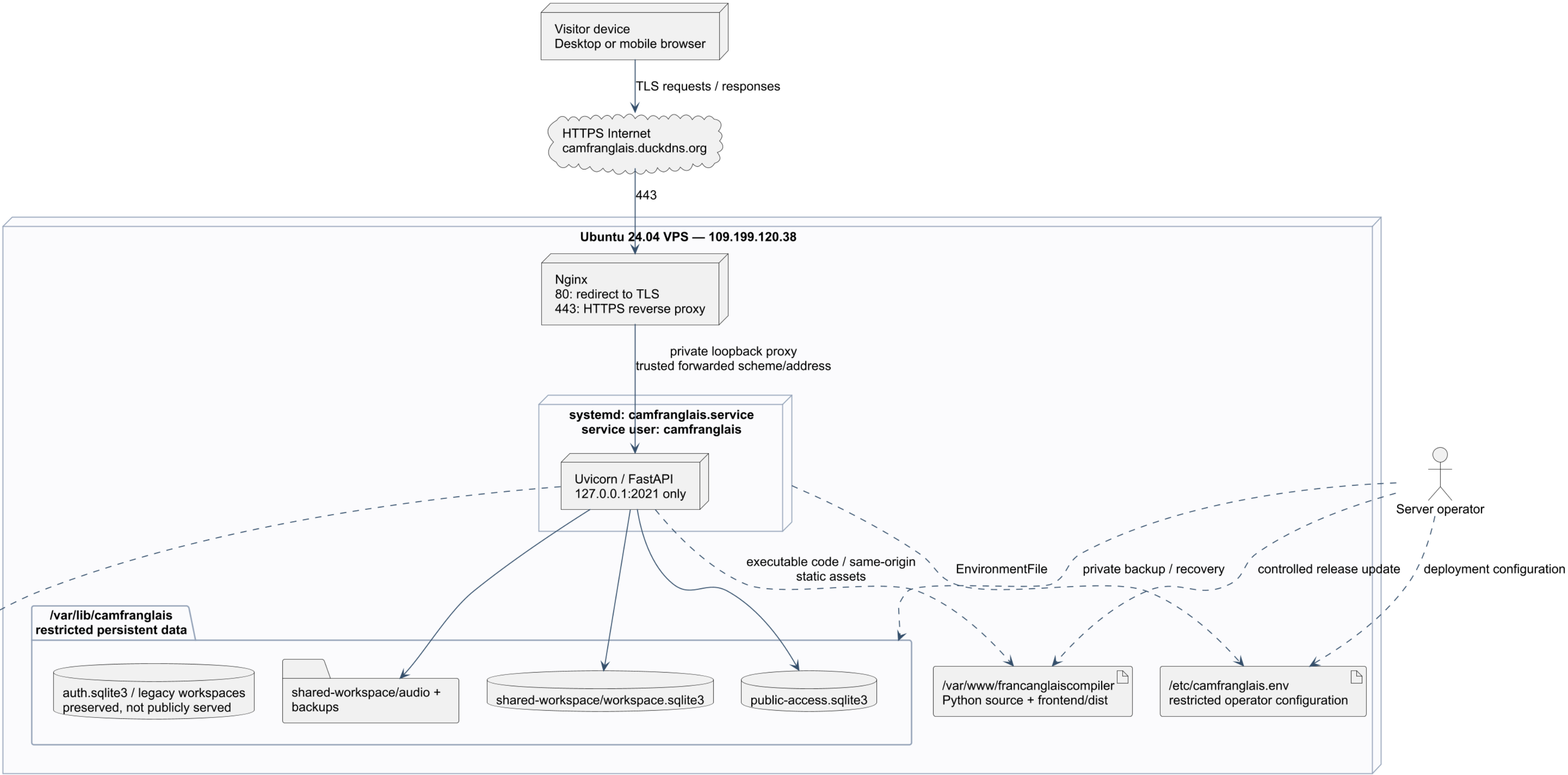


Camfranglais — public processing architecture



Components are modules, not invented runtime classes.
Only tests append; Collection, grammar and audio are read-only.
Operator backups/legacy APIs are outside this public UI.
No AI/voice generation, microphone or coursework export UI.

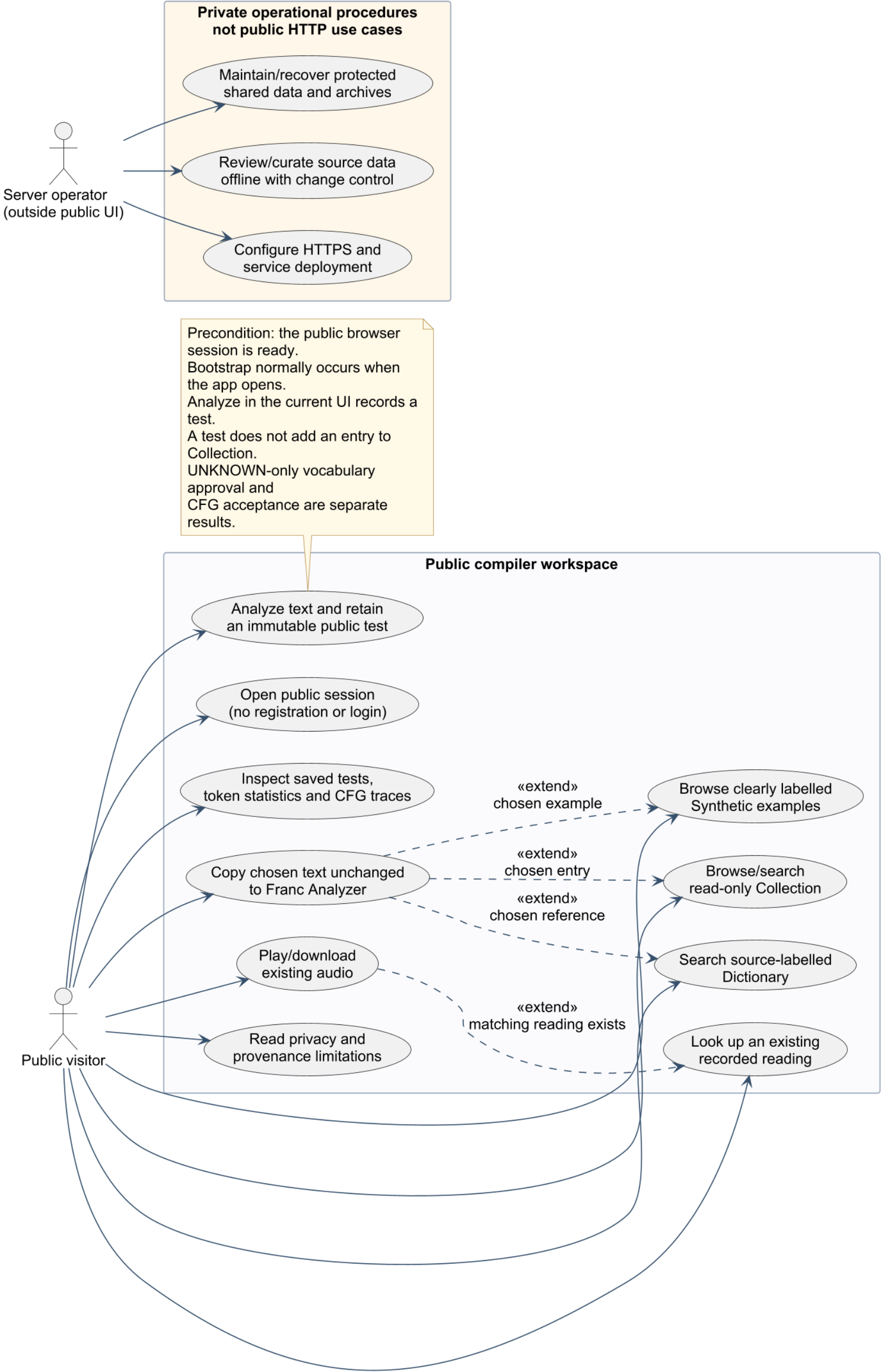
Camfranglais — target deployment and private persistence



Default release design:
public_read_only.
Production requires the built
frontend and a public HTTPS
origin.
SMTP and Google credentials are
not required in public mode.
Do not expose :2021 as a public
application URL.

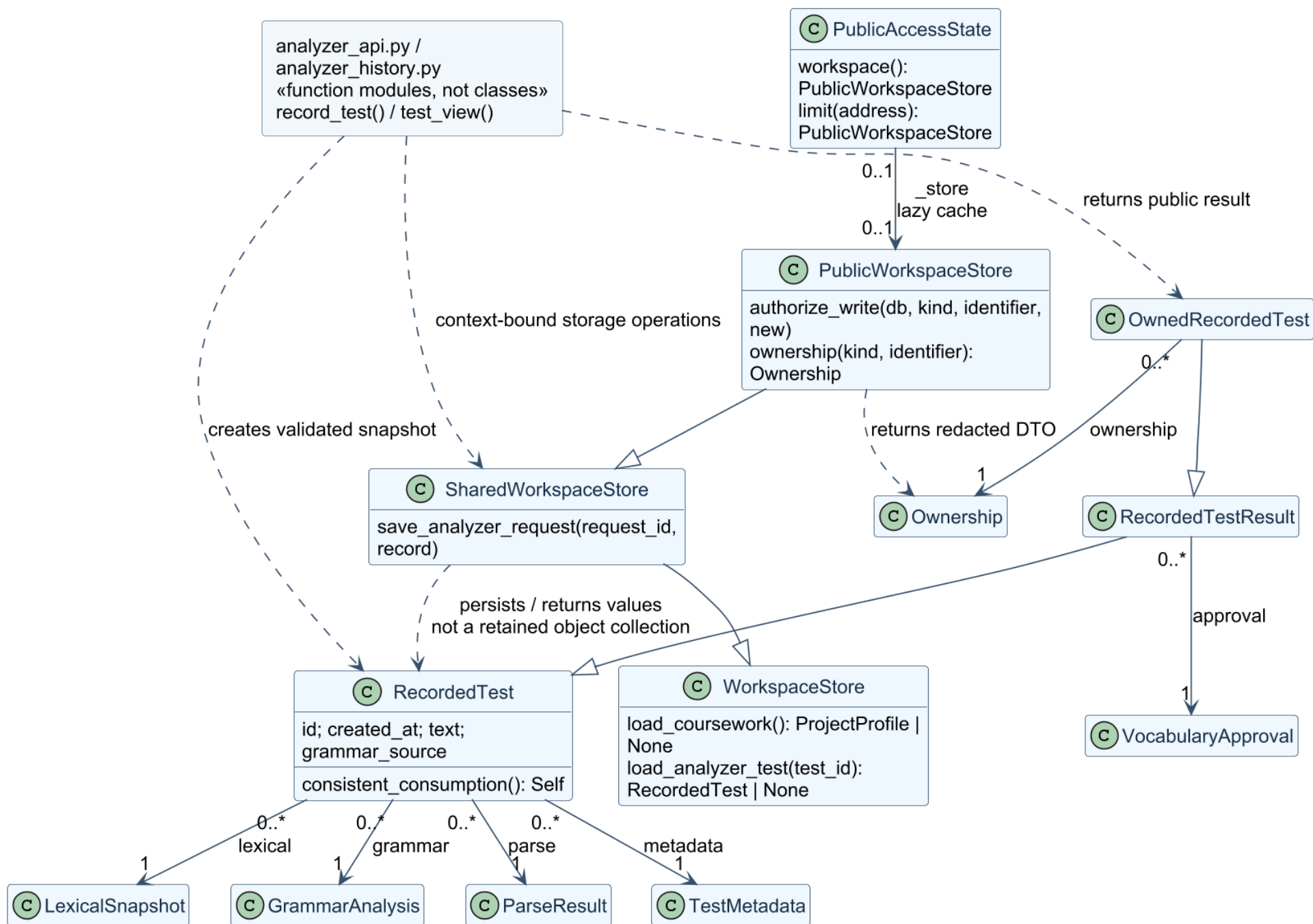
Intended internal host topology; the documentation session did not perform a remote update.
Read-only observation (27 September): bootstrap 200, public_read_only, five-page interface without login.
Browser/API observations do not certify exact server configuration, backup recovery or long-term uptime.
Operator recovery must preserve data/configuration; a workspace ZIP does not recreate account identities.

Camfranglais — public use cases, not account administration



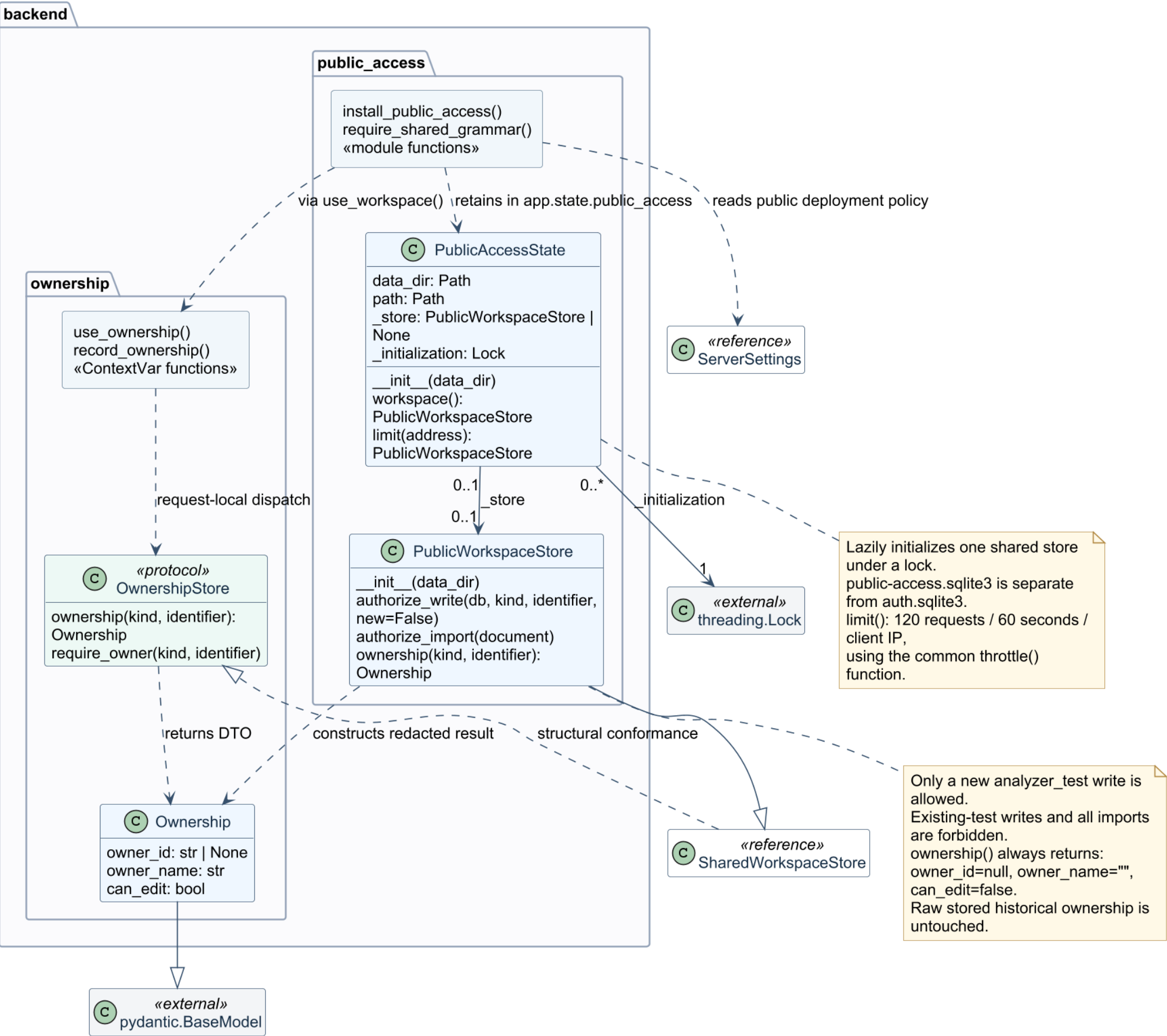
Read-only means protected reference resources; public test append is deliberately allowed.
No account controls, account counter, Collection/grammar/audio mutation or coursework export UI.
Recorded voice is playback, not browser synthesis, dictation or cloning.

Camfranglais — overall runtime relationships (orientation view)



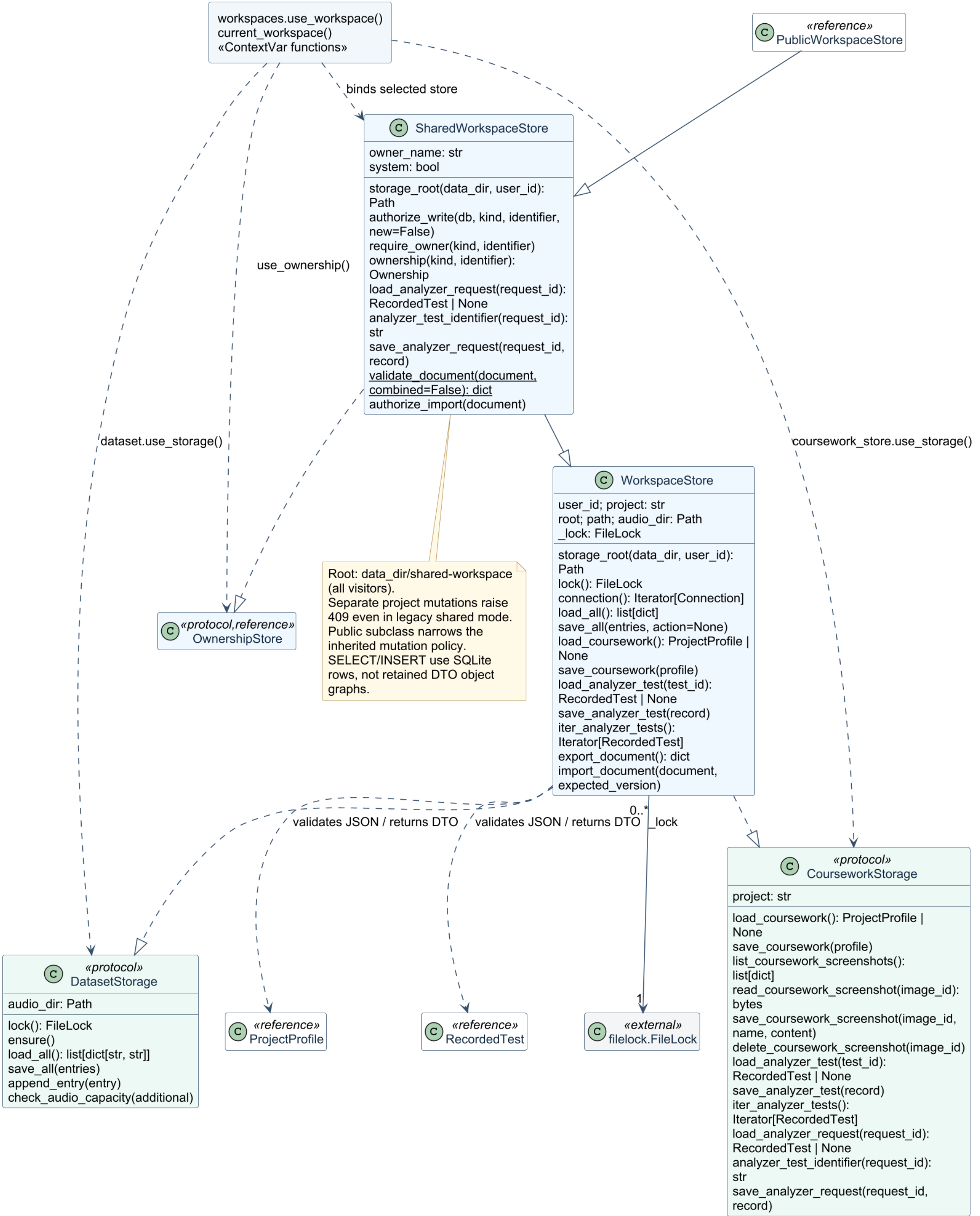
Orientation only: the detailed class sheets and class-coverage.json cover all 98 production classes. Hollow triangles: actual inheritance. Dashed arrows: usage dependencies, without multiplicities. Numbered arrows: actual stored/nested references; DTOs are not exclusive-lifecycle compositions. Public owner identity is redacted; the internal stable actor UUID is not a user account.

Public access — state, policy and ownership contracts



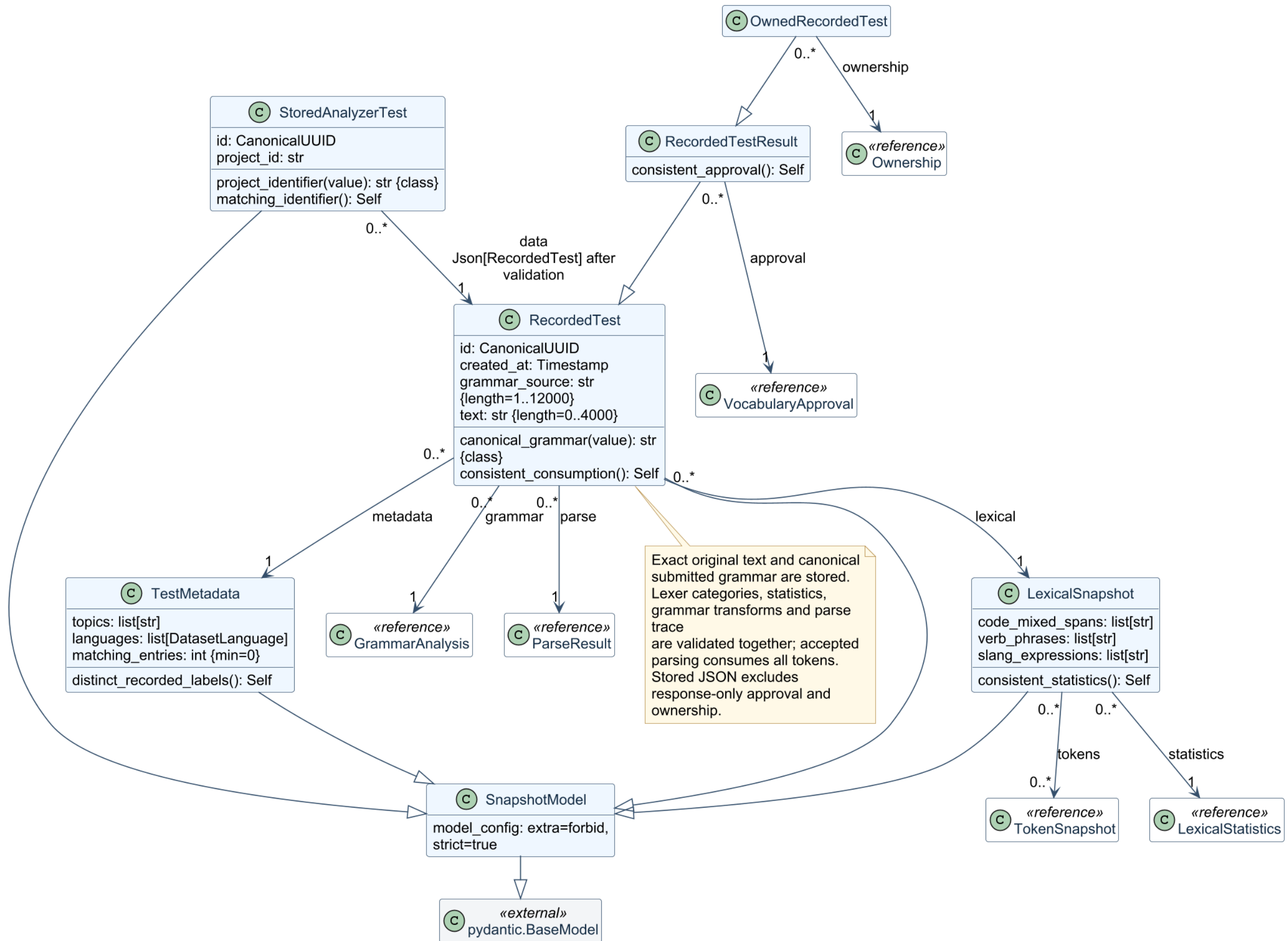
DatasetStorage and CourseworkStorage contracts are on class-workspaces. Stable PUBLIC_OWNER_ID derives from UUID5; it supports request-id lookup, not authentication. Opaque HttpOnly browser cookie + CSRF protect POST requests; no AuthStore or fake account is created.

Shared persistence — real store hierarchy and structural protocols



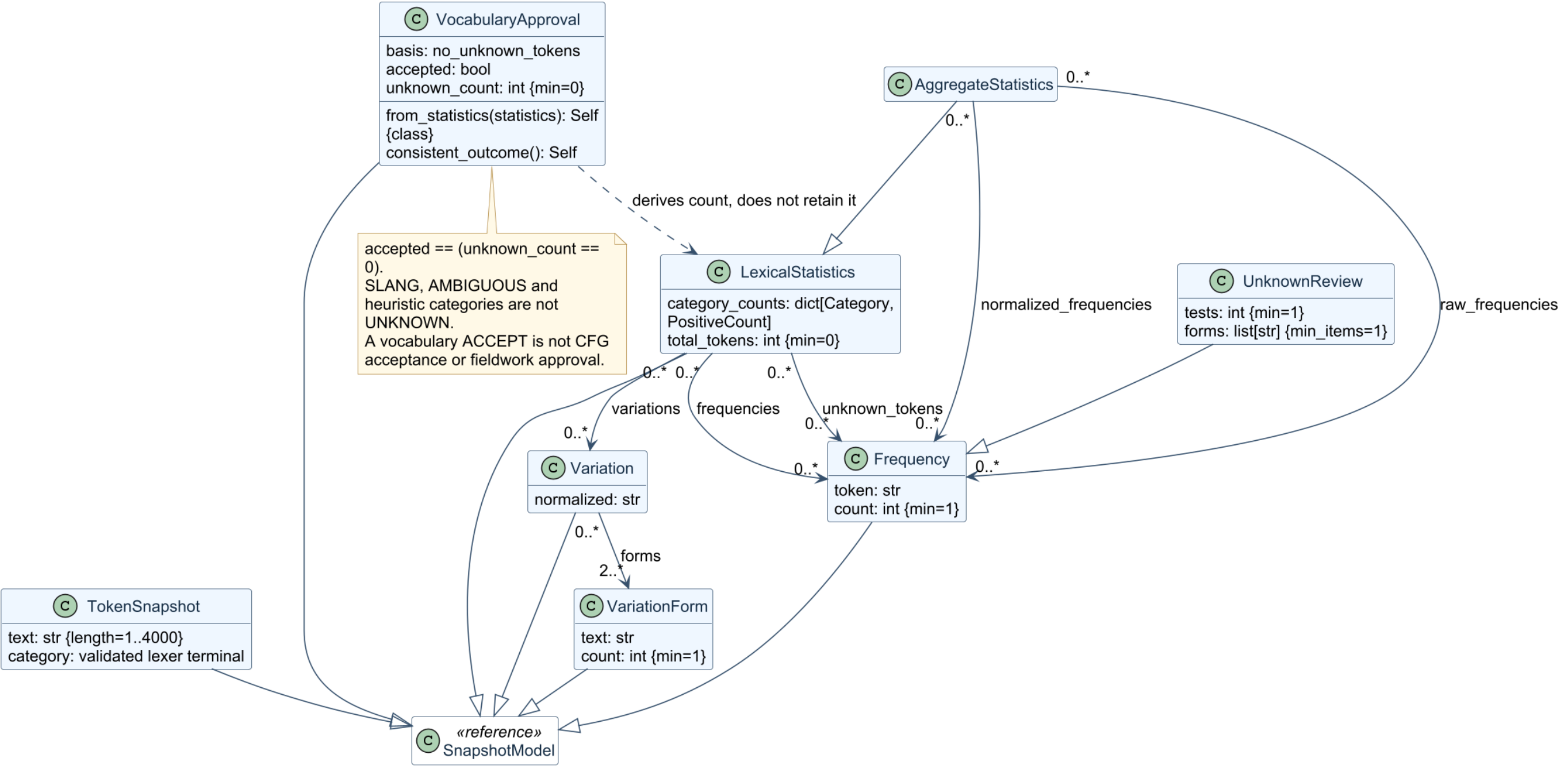
Hollow triangles toward bases; dotted triangles are structural Protocol conformance. WorkspaceStore also retains revision/history/screenshot operations for private compatibility. use_workspace() resets all request-local bindings on exit; no global file path is reassigned. A per-store file lock serializes related shared reads/writes; SQLite transactions protect logical rows.

Immutable tests — persisted snapshots and public result enrichment



Source: backend/analyzer_models.py. DTO serialization/validation also uses inherited Pydantic methods. Public results add derived vocabulary approval and a redacted ownership DTO without rewriting snapshots. Immutability is a public persistence policy, not Python frozen=True on these Pydantic models.

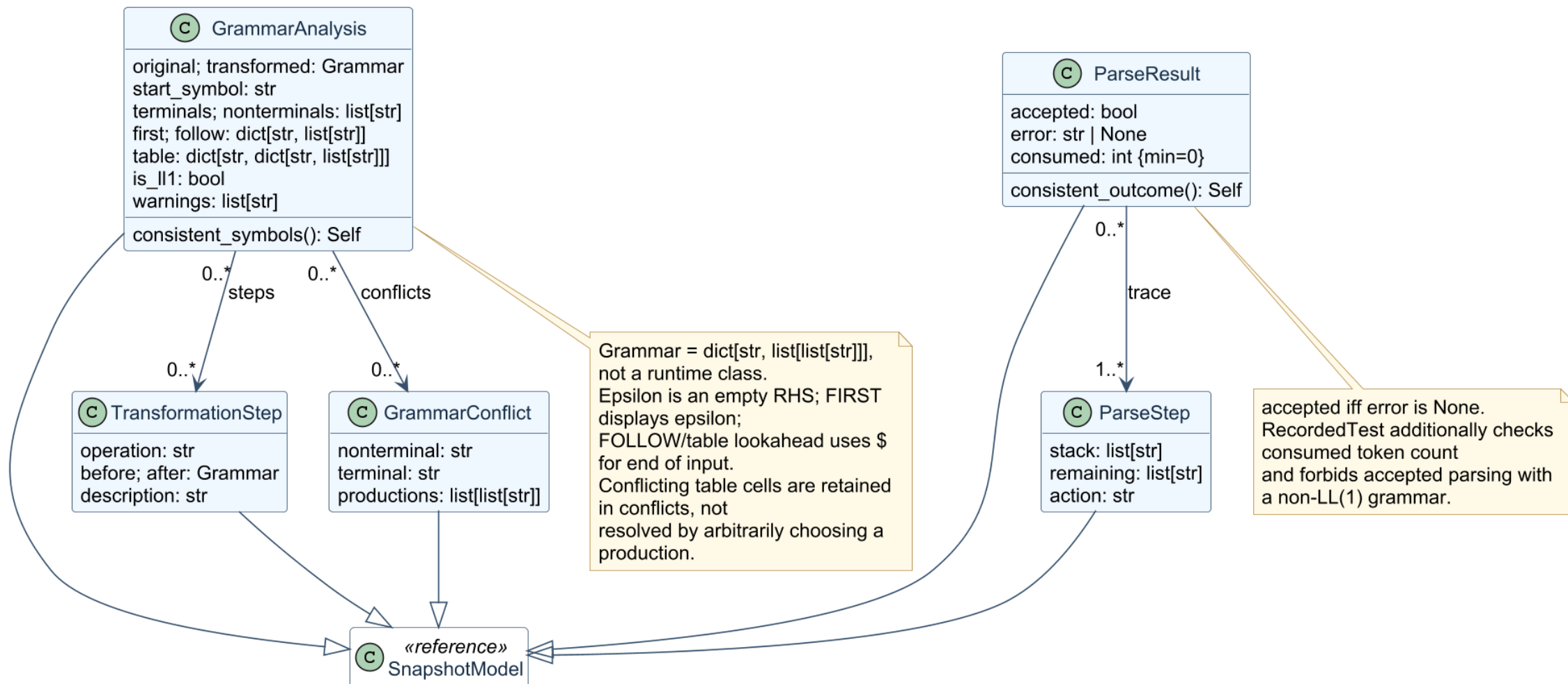
Lexical snapshots, token counts and UNKNOWN-only approval



accepted == (unknown_count == 0).
SLANG, AMBIGUOUS and heuristic categories are not UNKNOWN.
A vocabulary ACCEPT is not CFG acceptance or fieldwork approval.

Source: backend/analyzer_models.py. All displayed generalizations are actual base classes.
Token text retains spelling; casefolded and normalized frequency keys are separate projections.
Counts/variations are values and nested references, not a fabricated TokenStatistics service class.

Grammar analysis and predictive parsing — snapshot DTOs



Source: backend/analyzer_models.py. These are real DTO classes, not the compiler's internal dictionaries. Compiler functions create dictionaries; snapshot validation constructs these typed nested values. GrammarAnalysis and ParseResult are separately referenced by RecordedTest, not associated with each other.

Saved-test reporting — two verdict totals and paged summaries

analyzer_history.test_report(offset, limit)
 «function»
 streams every saved test, pages summaries

OwnedTestSummary

TestReport
 approval_basis:
 no_unknown_tokens
 topic_counts; language_counts:
 dict[str, PositiveCount]
 offset: int {min=0}
 limit: int {range=1..100}

TestSummary
 id: CanonicalUUID
 created_at: Timestamp
 text: str {length=0..4000}
 accepted: bool
 token_count; unknown_count: int {min=0}
 error: str | None
 grammar_accepted: bool
 grammar_error: str | None

TestTotals
 total; accepted; rejected: int {min=0}
 acceptance_rate: float | None {range=0..100}

«reference» AggregateStatistics

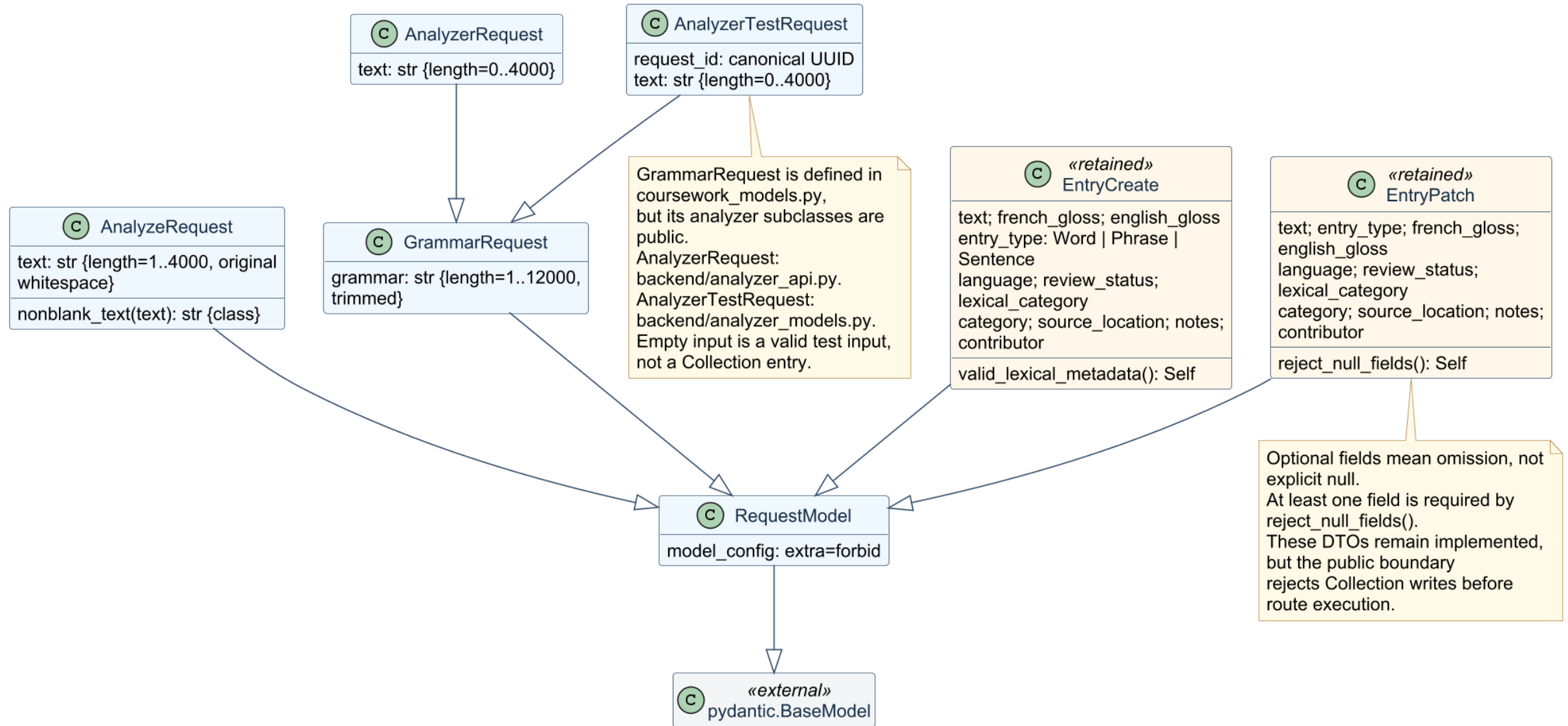
«reference» UnknownReview

«reference» SnapshotModel

Summaries are paged; totals/statistics use all retained tests. Repeated intentional tests remain separate observations. Collection counts are not this report's token statistics. Empty totals have acceptance_rate=null, not a fabricated percentage.

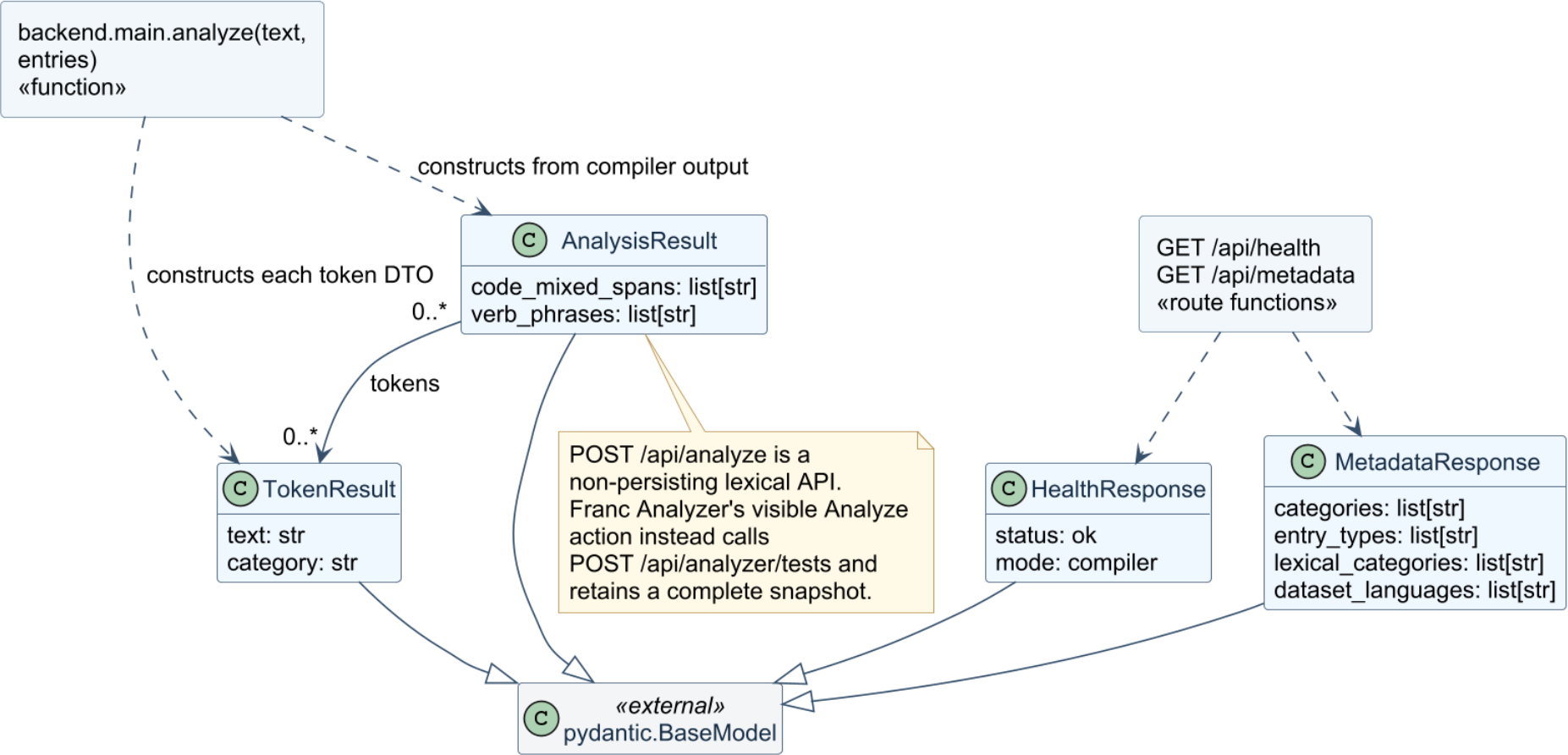
Source: backend/analyzer_models.py; aggregation: backend/analyzer_history.py.
 A report retains nested DTO references. SQL records are not model-object compositions.
 Public summaries redact ownership; vocabulary accepted/error and grammar_accepted/grammar_error stay distinct.

API inputs — current analyzer contracts and retained Collection mutations



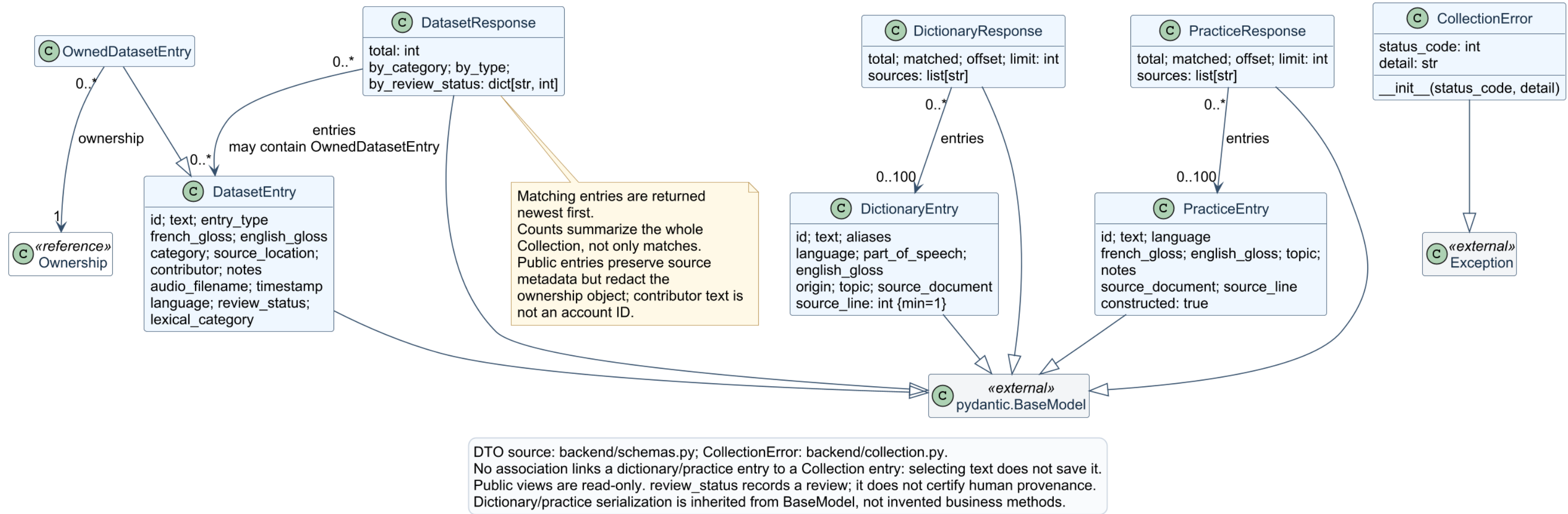
Other classes here: backend/schemas.py. Pydantic validation/serialization methods are inherited. The public analyzer uses the saved grammar; an arbitrary client grammar is not an editable public draft. Inheritance carries no multiplicities; no invented HTTP controller/request service class is introduced.

Small public API results — lexical preview, health and metadata

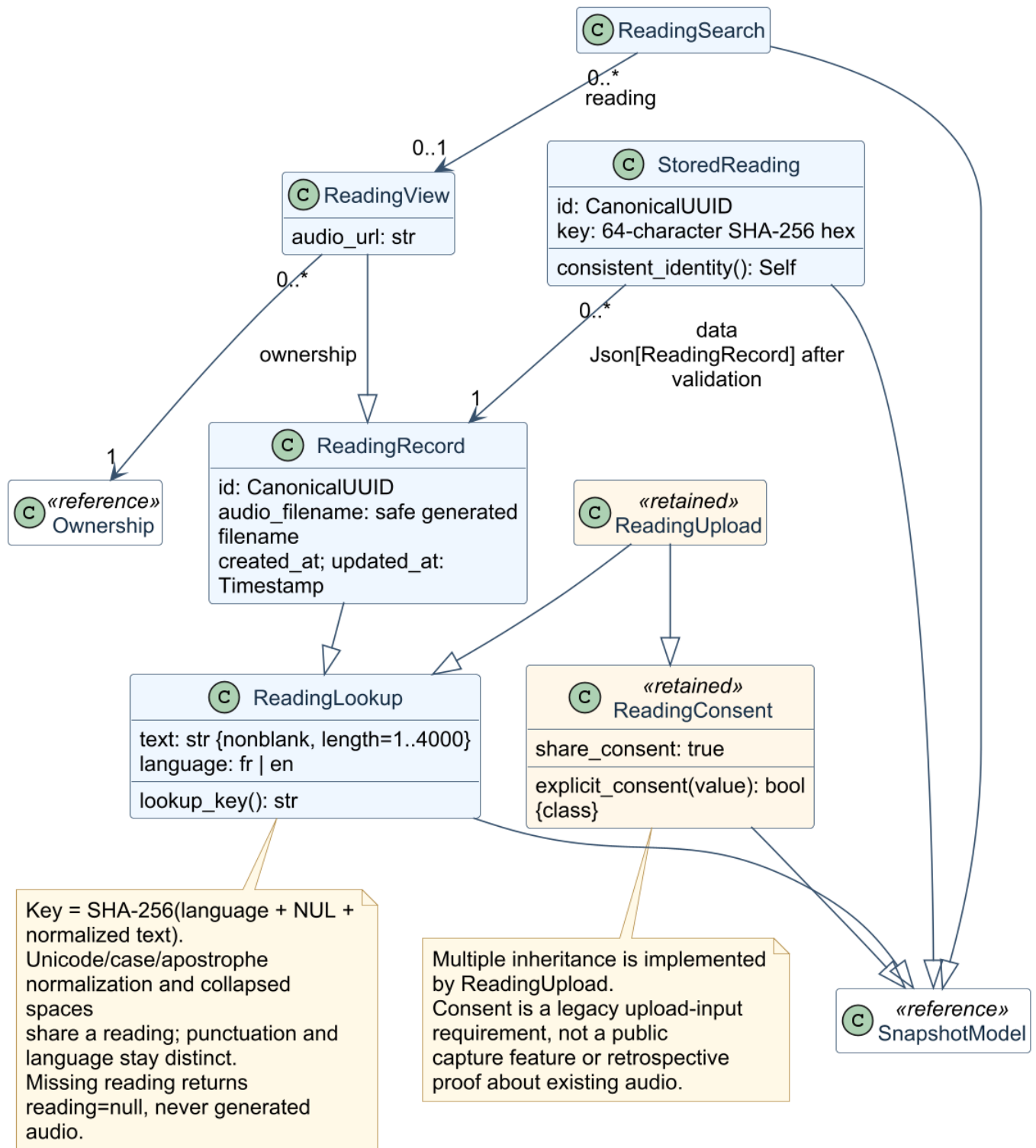


Source: backend/schemas.py. No selected authored methods: fields are the authored contract. BaseModel supplies serialization/validation; no invented constructors or business methods are listed. These DTOs are separate from strict TokenSnapshot and RecordedTest in analyzer_models.py.

Collection, Dictionary and Synthetic examples — separate data models



Recorded readings — public lookup/playback and retained upload contracts

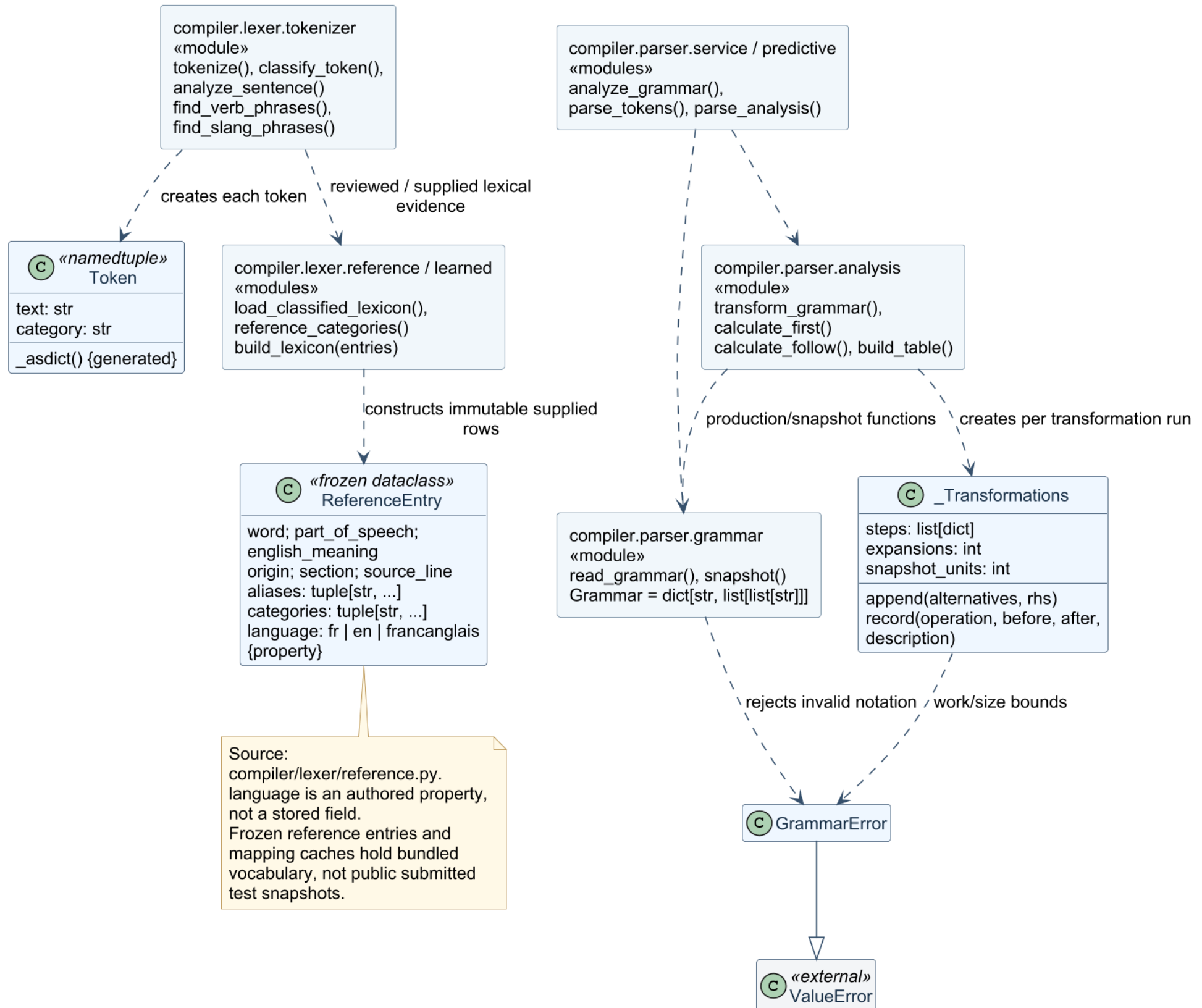


Key = SHA-256(language + NUL + normalized text).
Unicode/case/apostrophe normalization and collapsed spaces
share a reading; punctuation and language stay distinct.
Missing reading returns reading=null, never generated audio.

Multiple inheritance is implemented by ReadingUpload.
Consent is a legacy upload-input requirement, not a public capture feature or retrospective proof about existing audio.

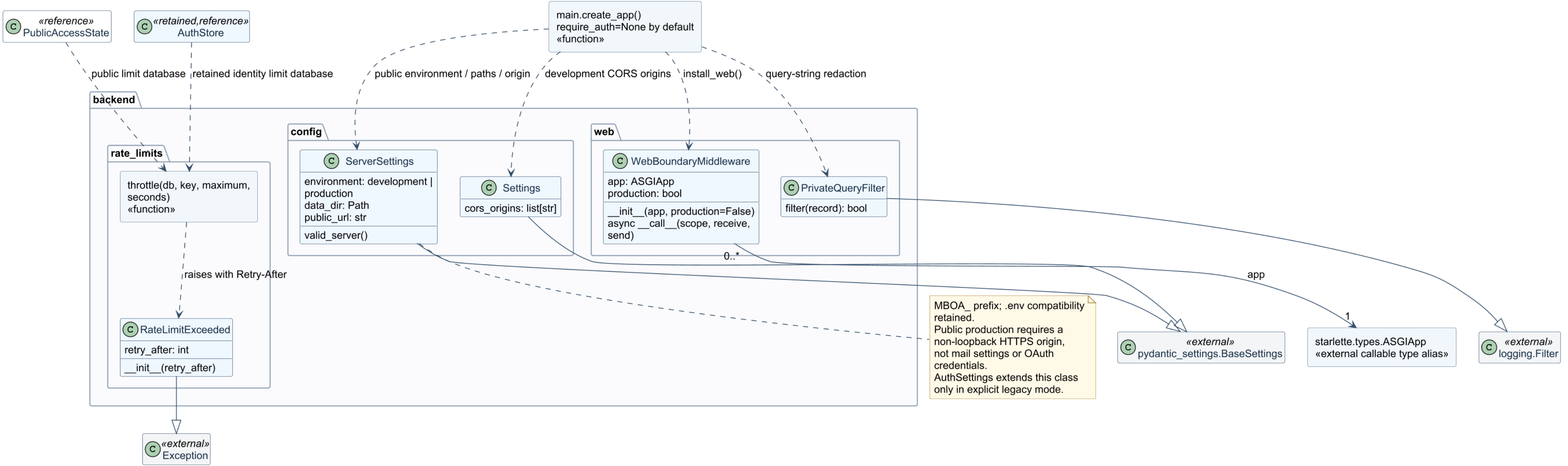
Source: backend/readings_models.py. Public routes: POST /api/readings/lookup; GET /api/readings/{id}/audio. Existing audio may be played/downloaded. Public create/replace/delete are denied. No browser TTS, microphone capture, automatic transcription or cloned voice is in this public workflow.

Compiler runtime — four authored classes, function-based algorithms



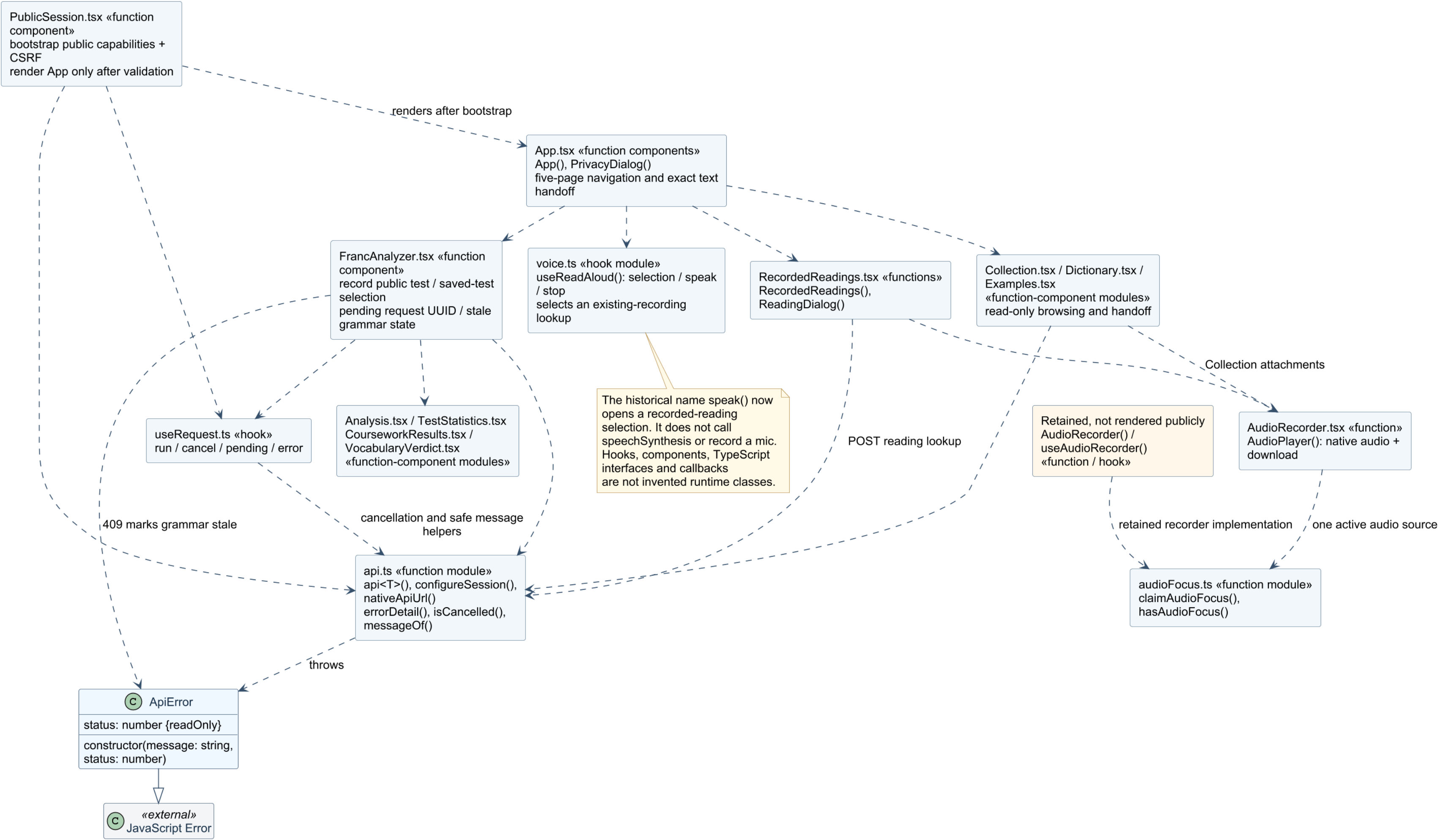
Token: compiler/lexer/tokenizer.py; GrammarError: compiler/parser/grammar.py;
_Transformations: compiler/parser/analysis.py. No LexerService, Parser or Grammar runtime class exists.
The general teaching starter has a JSON analysis cache; custom/saved corpus grammar is analyzed afresh.
Raw text stays intact; phrase annotations do not replace or collapse lexical tokens.

Runtime settings, web boundary and shared throttling



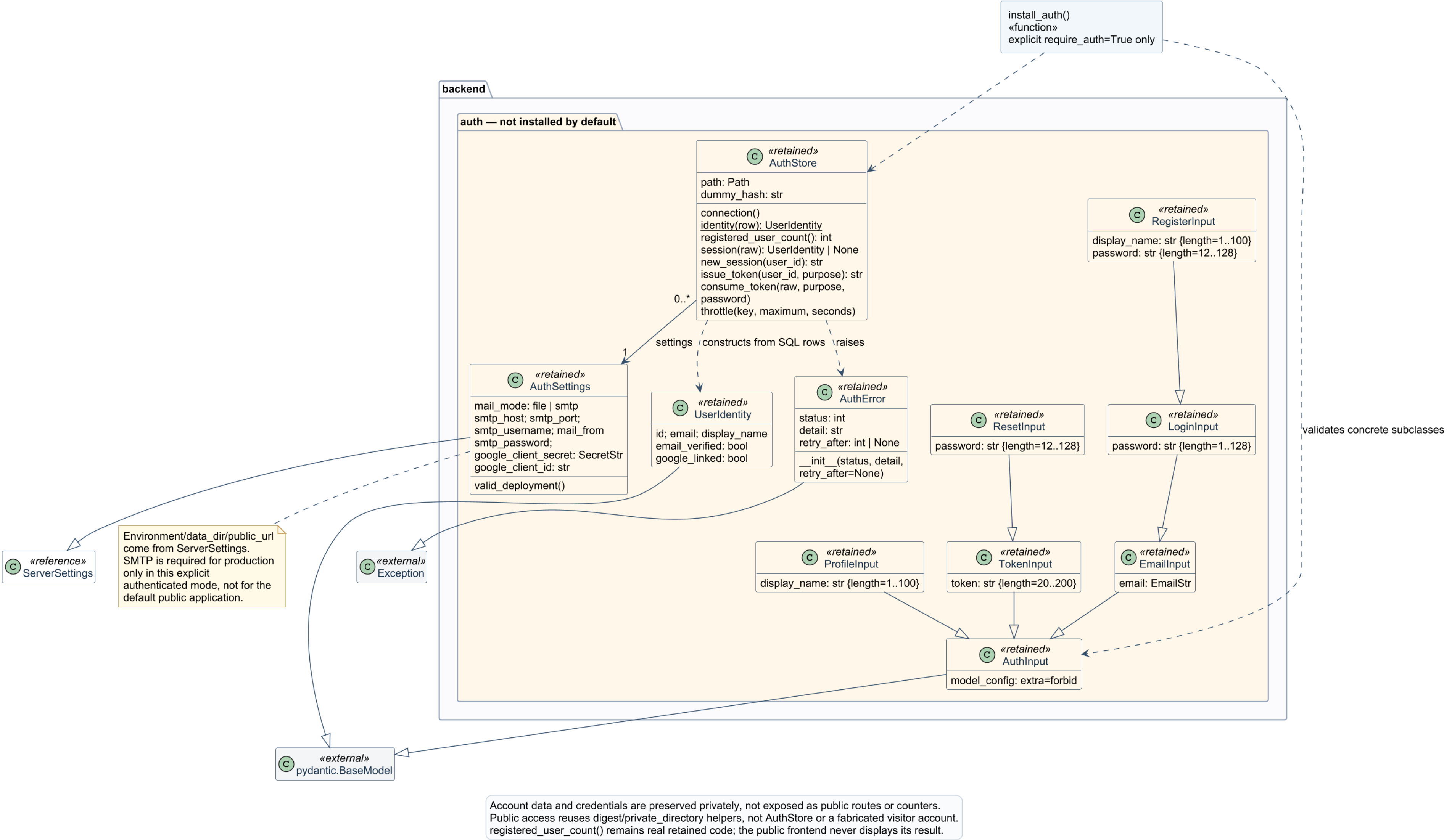
The ASGI reference is an external callable type alias, not an invented runtime class. Settings and ServerSettings are independent BaseSettings subclasses. Web boundary applies in all modes; public policy is in public_access.py. A microphone Permissions-Policy allowance does not create a public recording feature.

React implementation — one authored class, functions and hooks



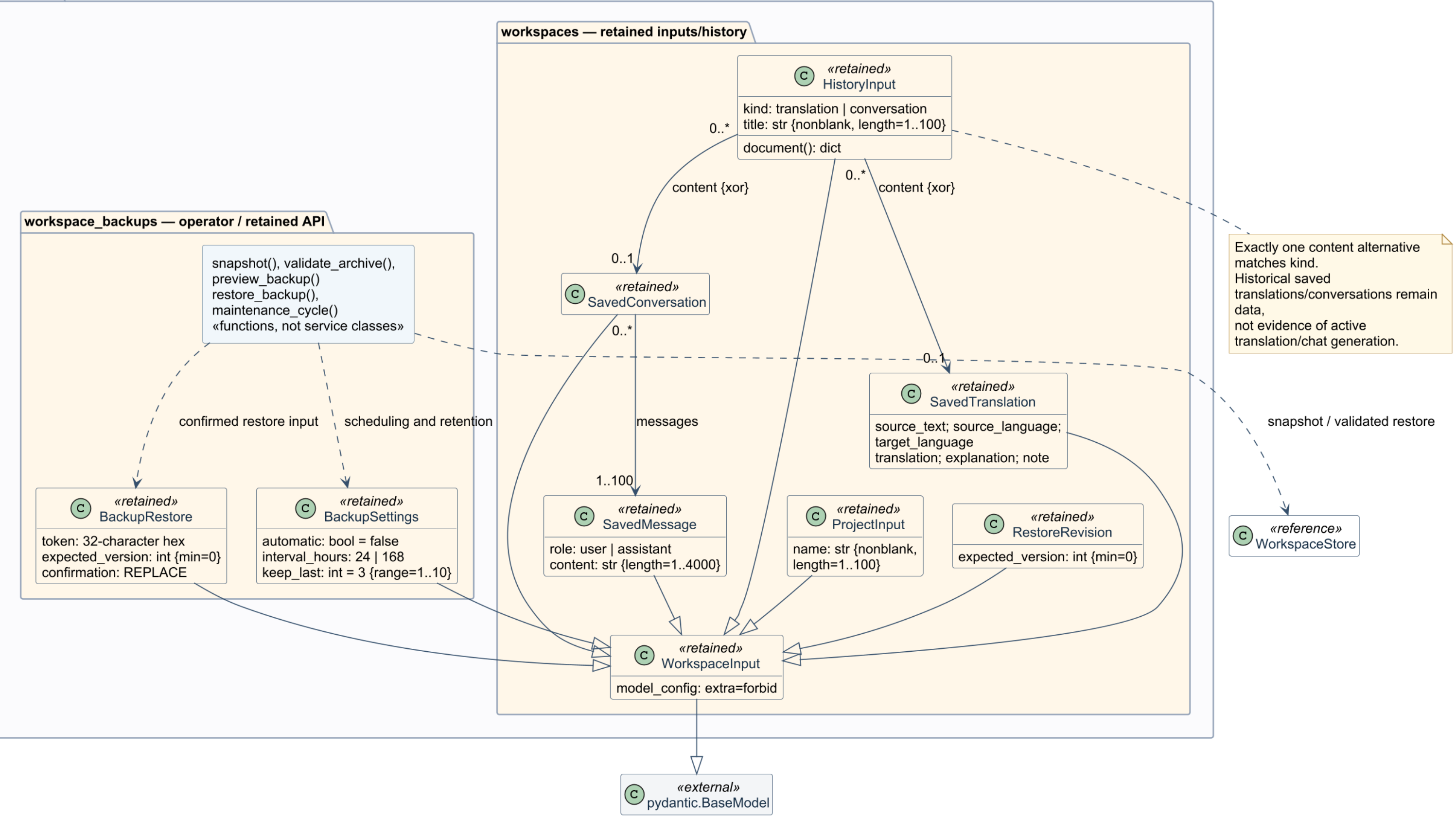
AuthGate, account controls, project selectors and account counter are absent from the current frontend. Only frontend.src.api.ApiError is an authored production TypeScript class. Existing AudioRecorder/useAudioRecorder source does not imply a public capture/upload capability.

Retained accounts — explicit legacy mode only



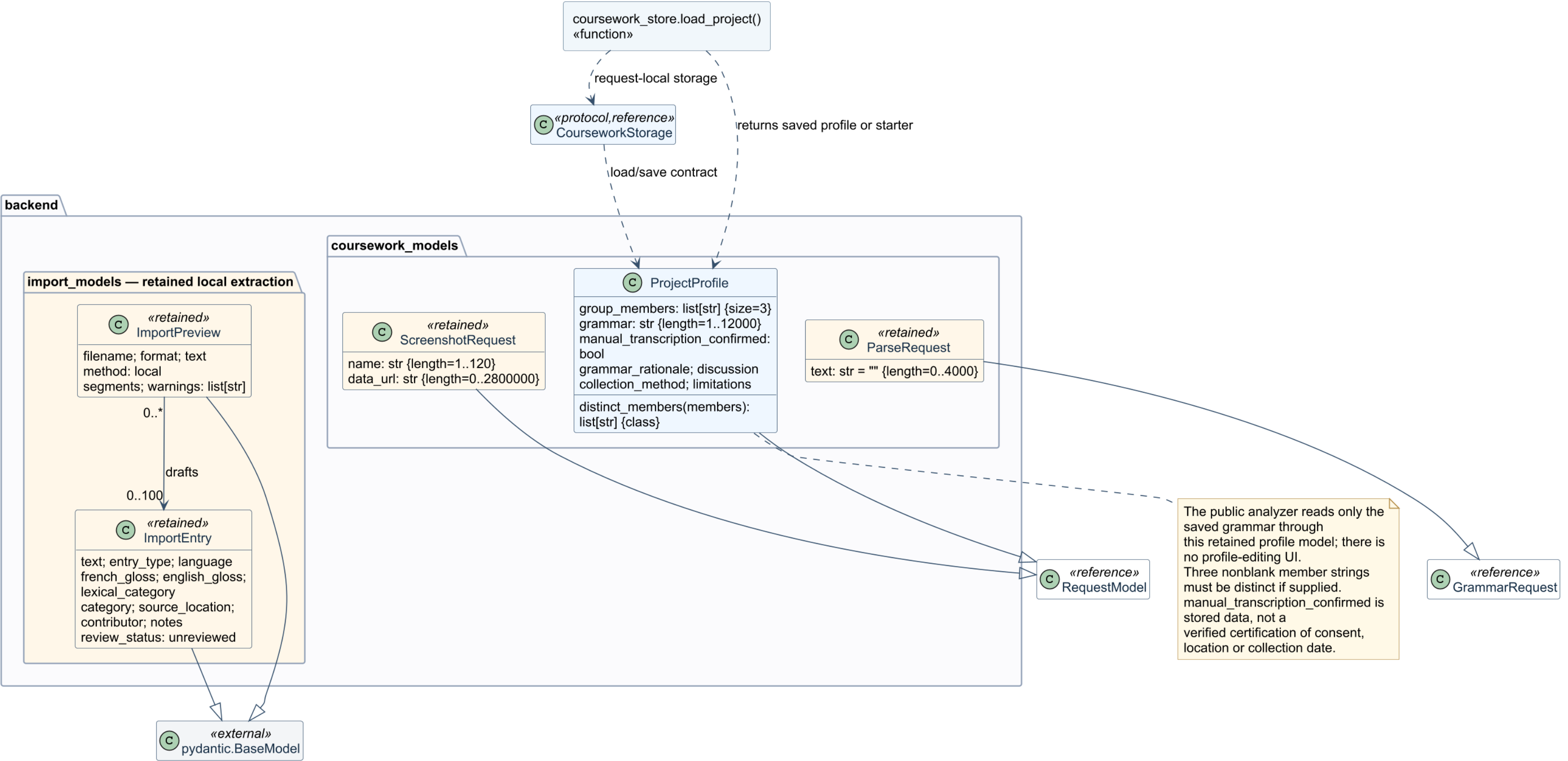
Retained workspace inputs, old saved history and operator backups

backend



All workspace/history/import/backup HTTP endpoints are unavailable in the default public application. Automatic backup maintenance may still run privately when configured; no public recovery UI is mounted. Shared export is format 5 (format 4 compatible); private WorkspaceStore export is format 3. Older private formats 1/2 normalize during validation. ZIP manifest remains format 1.

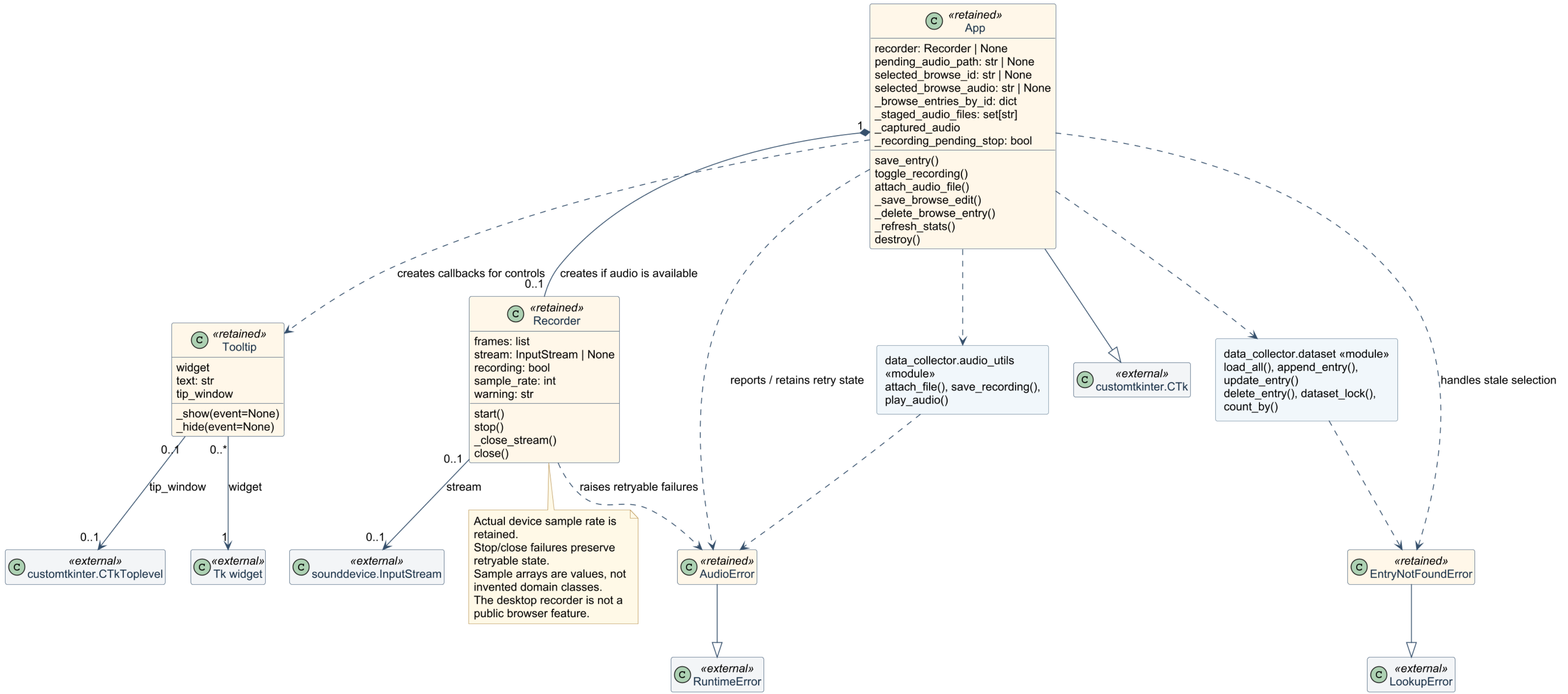
Retained coursework/import contracts — saved grammar reuse, no public export UI



The public analyzer reads only the saved grammar through this retained profile model; there is no profile-editing UI. Three nonblank member strings must be distinct if supplied. `manual_transcription_confirmed` is stored data, not a verified certification of consent, location or collection date.

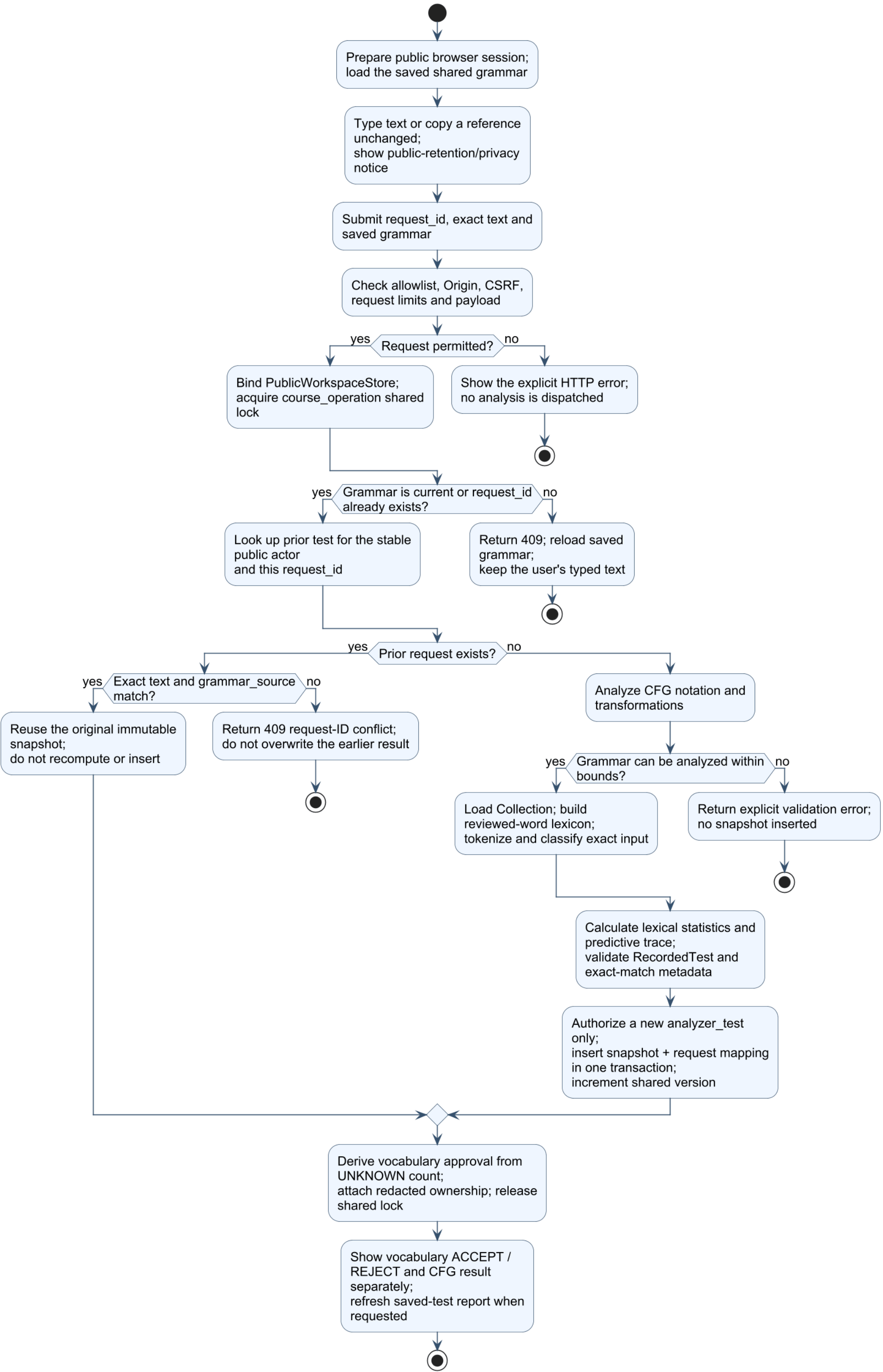
ProjectProfile still participates in active grammar loading; the other authored classes here are retained compatibility. GrammarRequest is a shared public analyzer input base, detailed in class-api-requests. Import preview, screenshots, coursework editing and generated report/presentation routes are not mounted publicly. Profile/screenshot/archive preservation does not expose legacy data through the default public application.

Retained desktop collector — actual GUI, helper and recorder classes



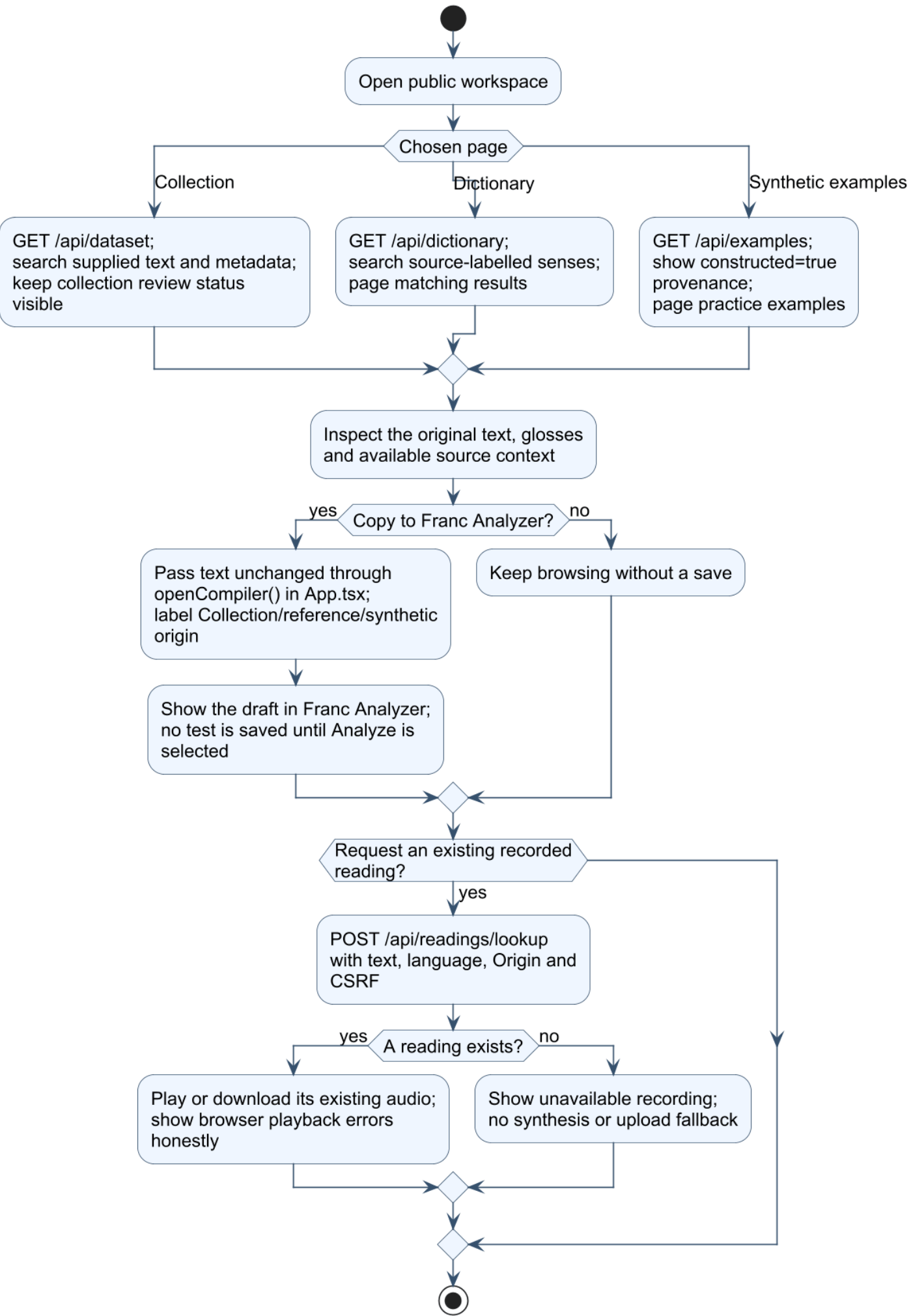
App/Tooltip: data_collector/App.py; Recorder/AudioError: audio_utils.py; EntryNotFoundError: dataset.py.
App creates the optional Recorder; bound widget callbacks retain Tooltip objects, not an App tooltip collection.
CSV and local capture workflows are retained operator/desktop compatibility, not the default web product.

Public Analyze — saved grammar, immutable tests and separate verdicts



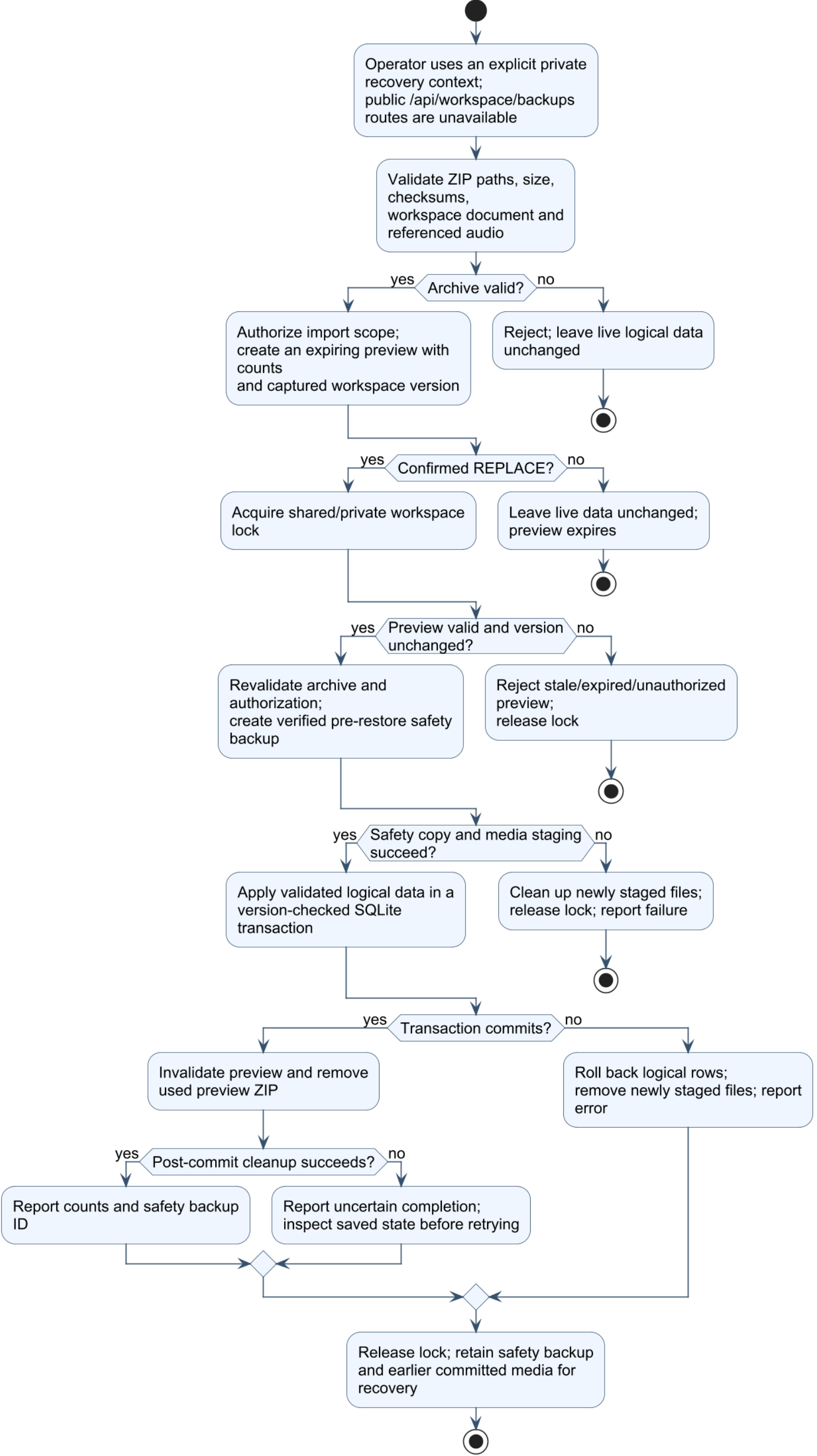
Analyze in the UI records a public test; Collection is never mutated by this workflow. A missing CFG table cell, non-LL(1) grammar or parser bound can yield a retained rejected test. Stopping client-side waiting does not cancel or roll back a server commit; inspect saved tests before retry. An existing identical retry may return its old snapshot even after offline grammar curation.

Read-only reference use — browse, inspect and hand off without mutation



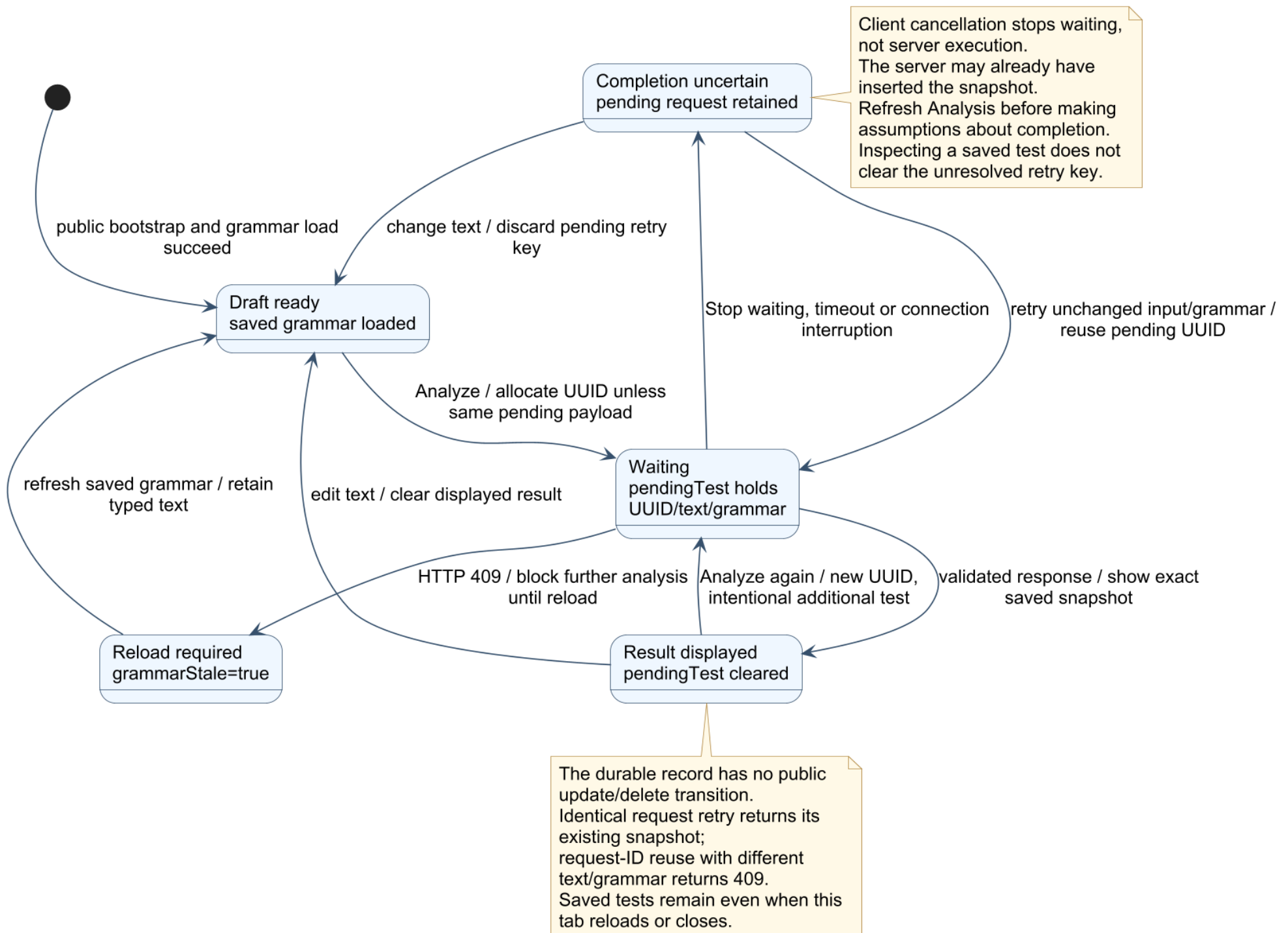
The sources are separate datasets, not subclasses of one collected-statement entity.
Human collection place/date/consent are unconfirmed; a label or review flag is not fieldwork certification.
No Collection, grammar or audio editing occurs; browsing and handoff never add an entry or a test.

Retained restore procedure — private operator/legacy workflow only



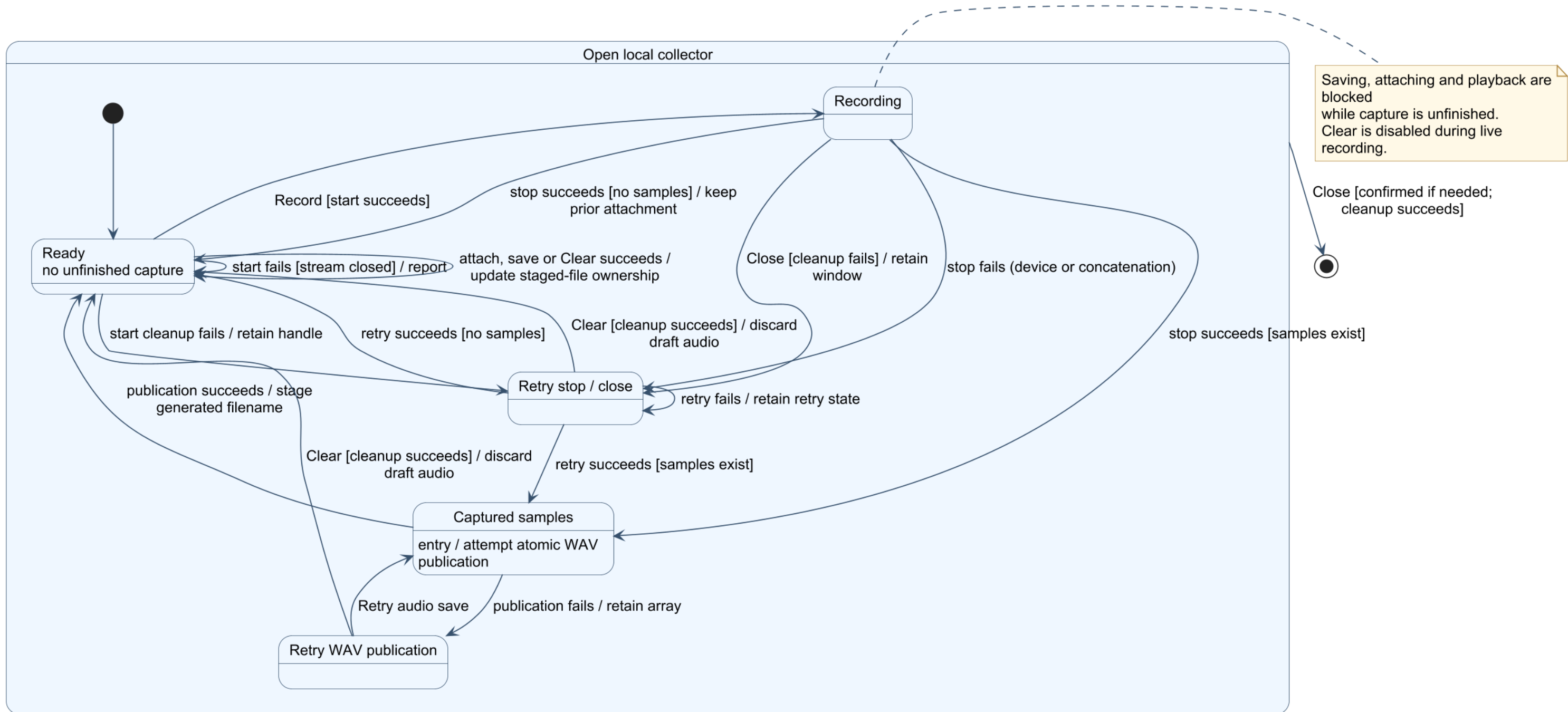
Shared format 5 includes ownership, request mappings, archived profiles and readings; format 4 remains compatible. Private formats 1/2 normalize to format 3; outer ZIP manifest is format 1. Non-system legacy restore cannot change another creator's records or original archived profiles. PublicWorkspaceStore.authorize_import() always rejects; no browser restore feature is implied. Filesystem staging and SQLite are coordinated operations, not one cross-file ACID transaction.

Public test submission — browser state versus durable result



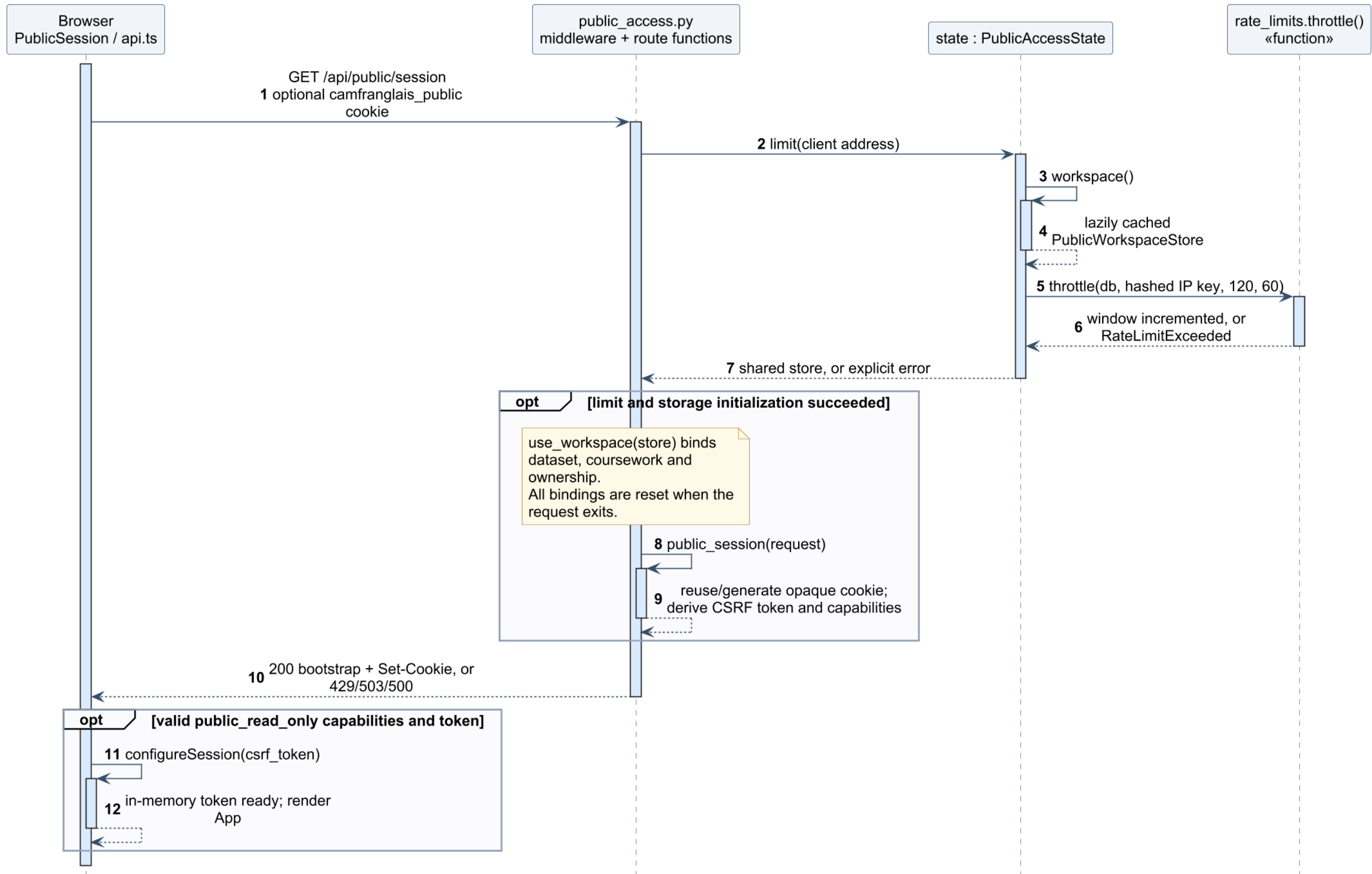
Derived logical browser states, not an invented enum or application class.
Source: frontend/src/FrancAnalyzer.tsx, api.ts, useRequest.ts; persistence: analyzer_history.py/public_access.py.
Selecting an old snapshot does not rerun it against today's Collection or grammar.

Retained desktop only — recorder readiness and retry states



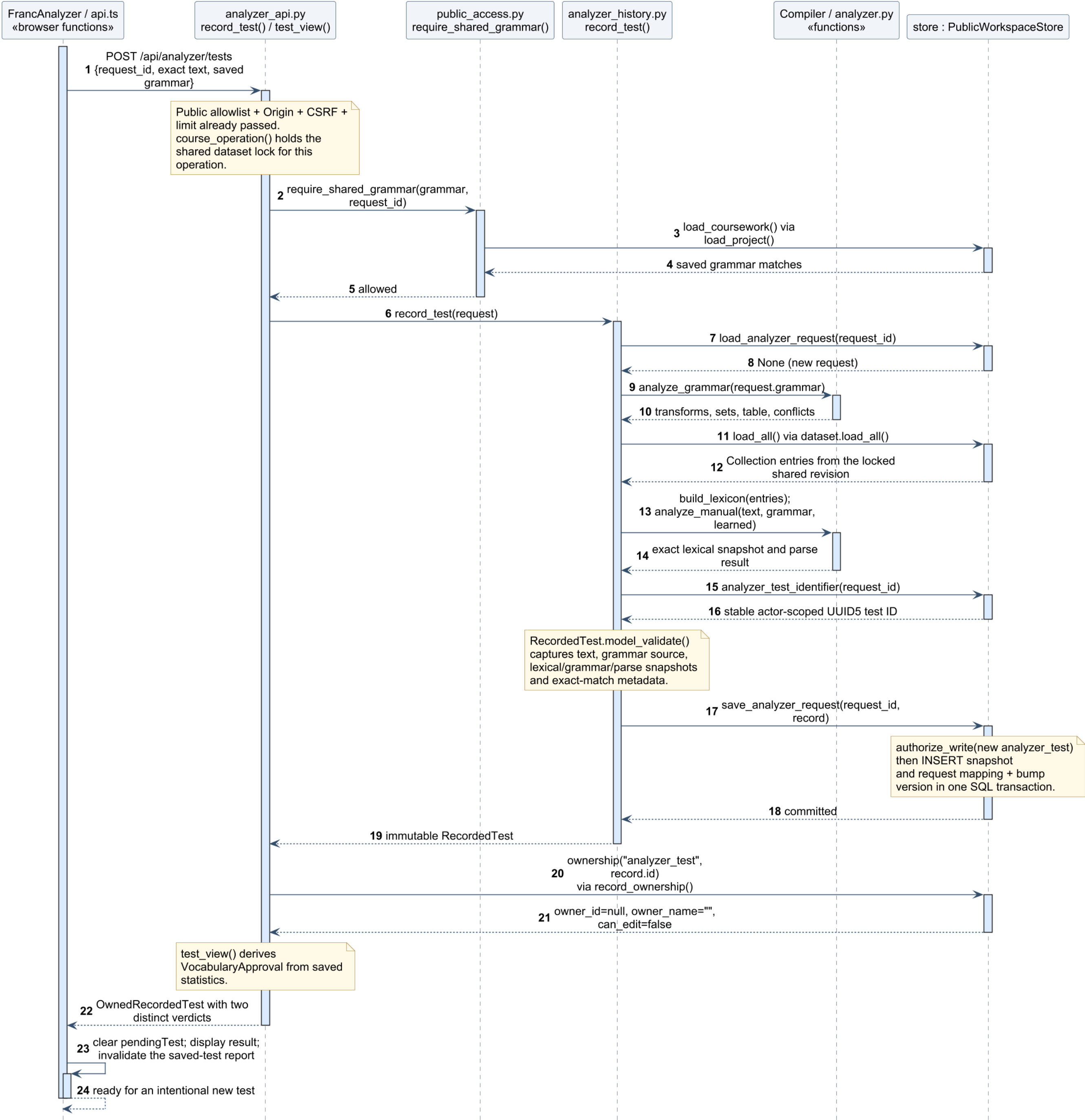
Source: `data_collector/App.py` and `audio_utils.py`. This is not a public browser microphone feature. Logical states derive from recording, stream, pending-stop and captured-array fields, not invented classes/enums. Ready may include a staged filename. Cleanup failures preserve the retry state rather than pretending to close. Draft-owned files may be discarded; committed recordings are not owned by the draft.

Public bootstrap — CSRF preparation without an account



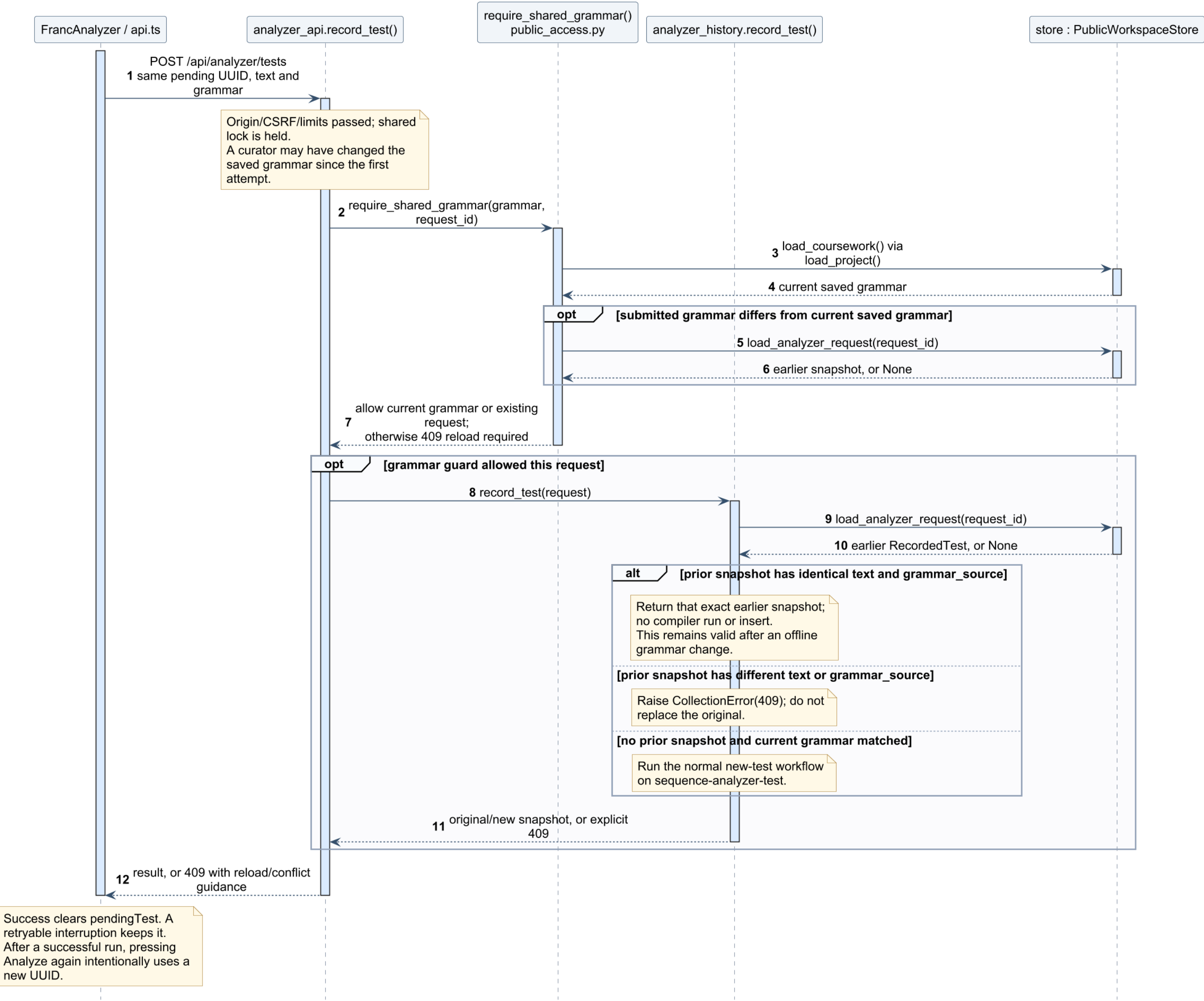
Successful capabilities: analyze/save_tests=true; edit_collection/edit_grammar/edit_recordings=false.
Cookie: HttpOnly, SameSite=Lax, Secure in production. It is CSRF state, not login or user identity.
Rate-limit rows live in public-access.sqlite3; no auth.sqlite3 initialization or fake account is required.
Health bypasses the shared limit; this sequence shows the default include_academic=true bootstrap.

Analyze and retain one new public test — successful path



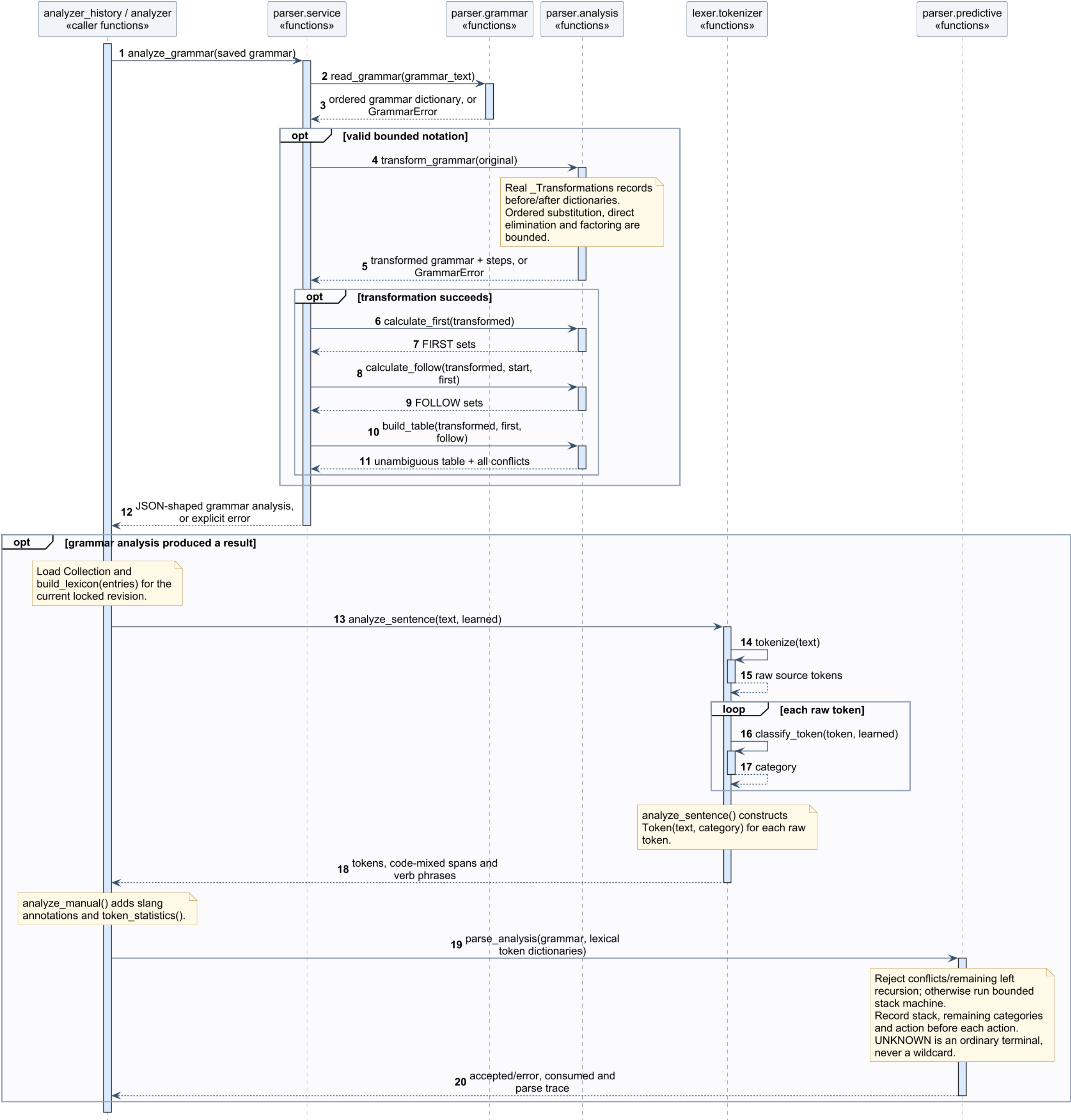
Storage calls abbreviate the real dataset/coursework_store/ownership ContextVar adapters. A retained grammar rejection is still a completed test; malformed/oversized grammar raises before insertion. No Collection entry, grammar, recording or private history is modified by this operation. Cancellation/timeout cannot promise rollback; identical retries are detailed separately.

Test retry — preserved snapshot versus stale/new request



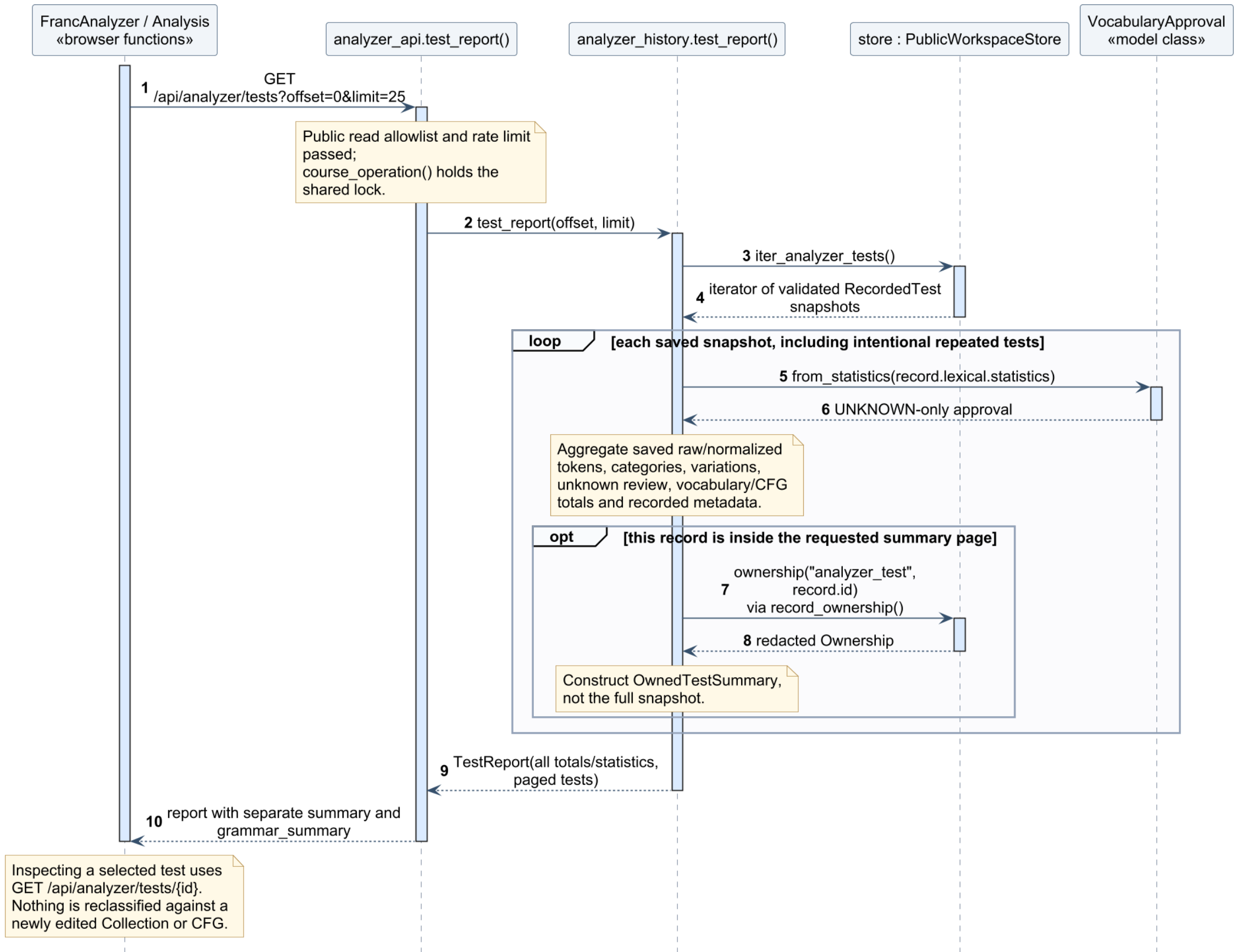
Stable PUBLIC_OWNER_ID makes retry lookup survive cookie changes; it is not a fabricated account. Public request IDs are idempotency keys, not credentials. Exact payload comparison prevents key reuse from overwriting data. A new stale request fails before analysis; existence alone does not bypass the later exact-payload check.

Deterministic compiler — actual function calls and trace construction



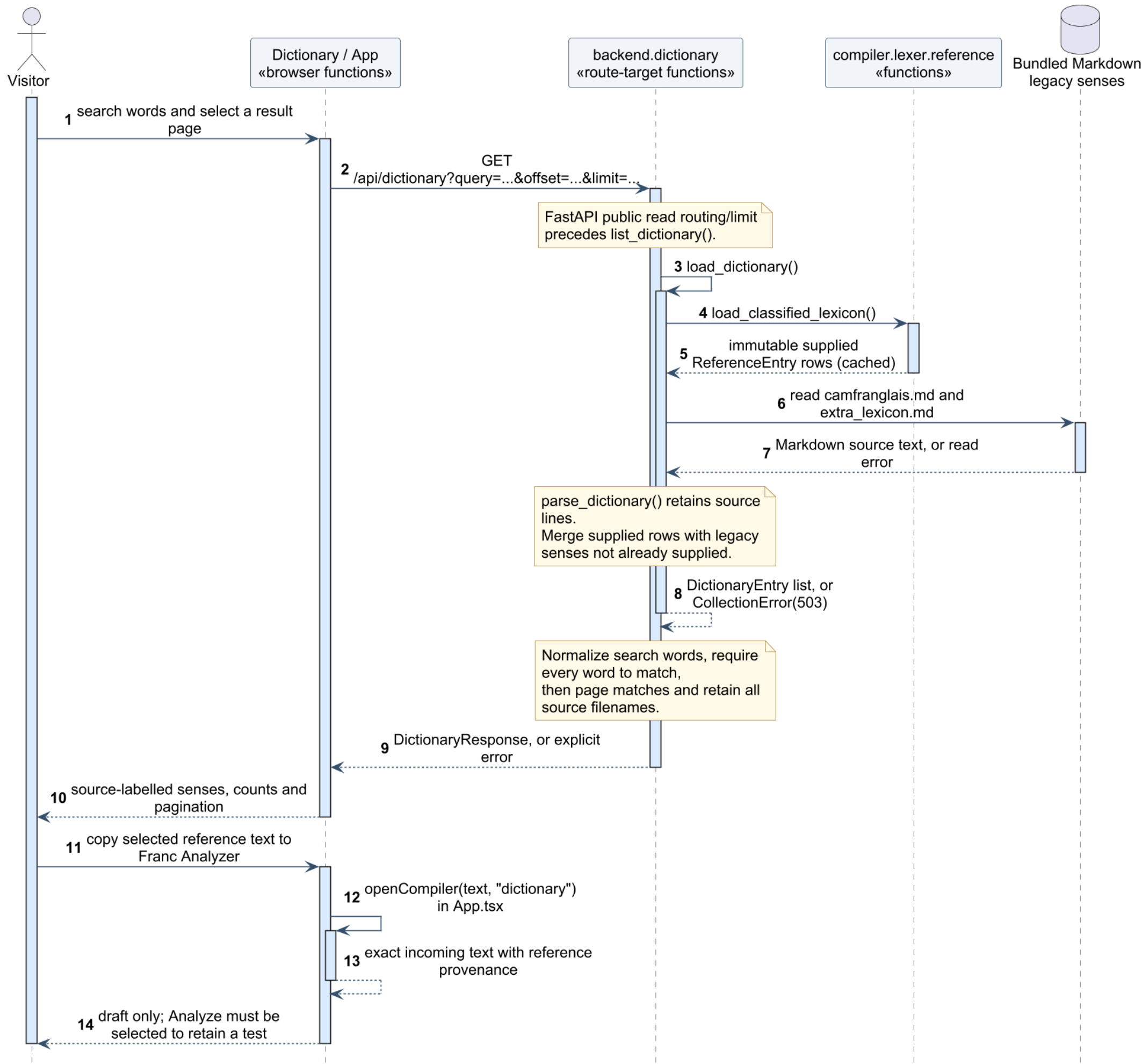
This shows the saved/custom grammar path; the generic teaching starter alone has a decoded JSON cache. Parser bounds: 256 input tokens, 1000 characters/token, 2000 trace steps, 1024 stack symbols. Vocabulary approval is a separate later calculation: only UNKNOWN causes that verdict to reject.

Analysis statistics — all saved snapshots, one page of summaries



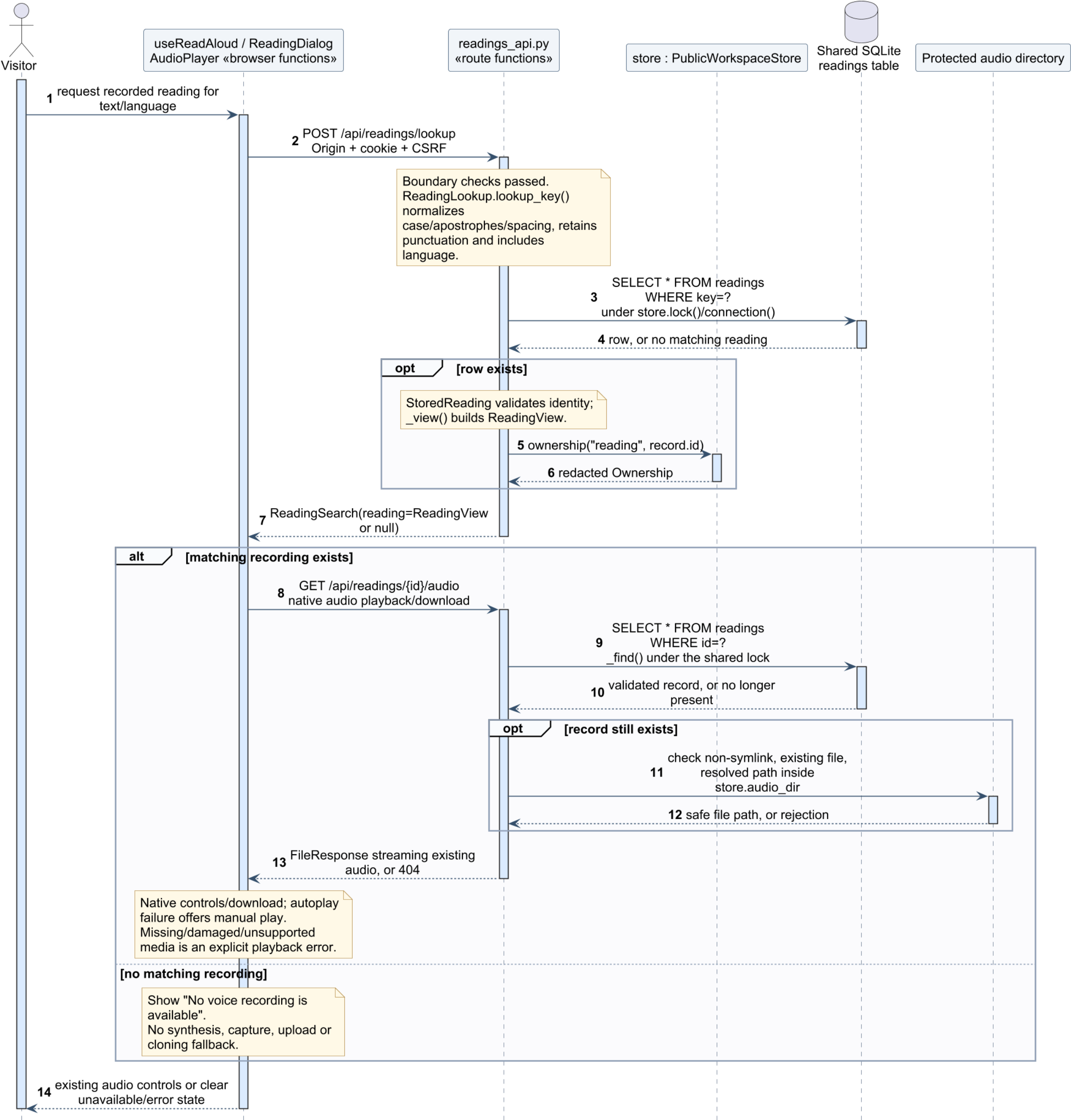
Iteration is lazy: the store validates SQL rows as consumed, retaining one complete snapshot at a time. offset/limit page only summaries, not the population underlying the aggregates. Empty acceptance rates are null; historical test counts are dated observations, not permanent constants.

Dictionary search and exact text handoff — no Collection write



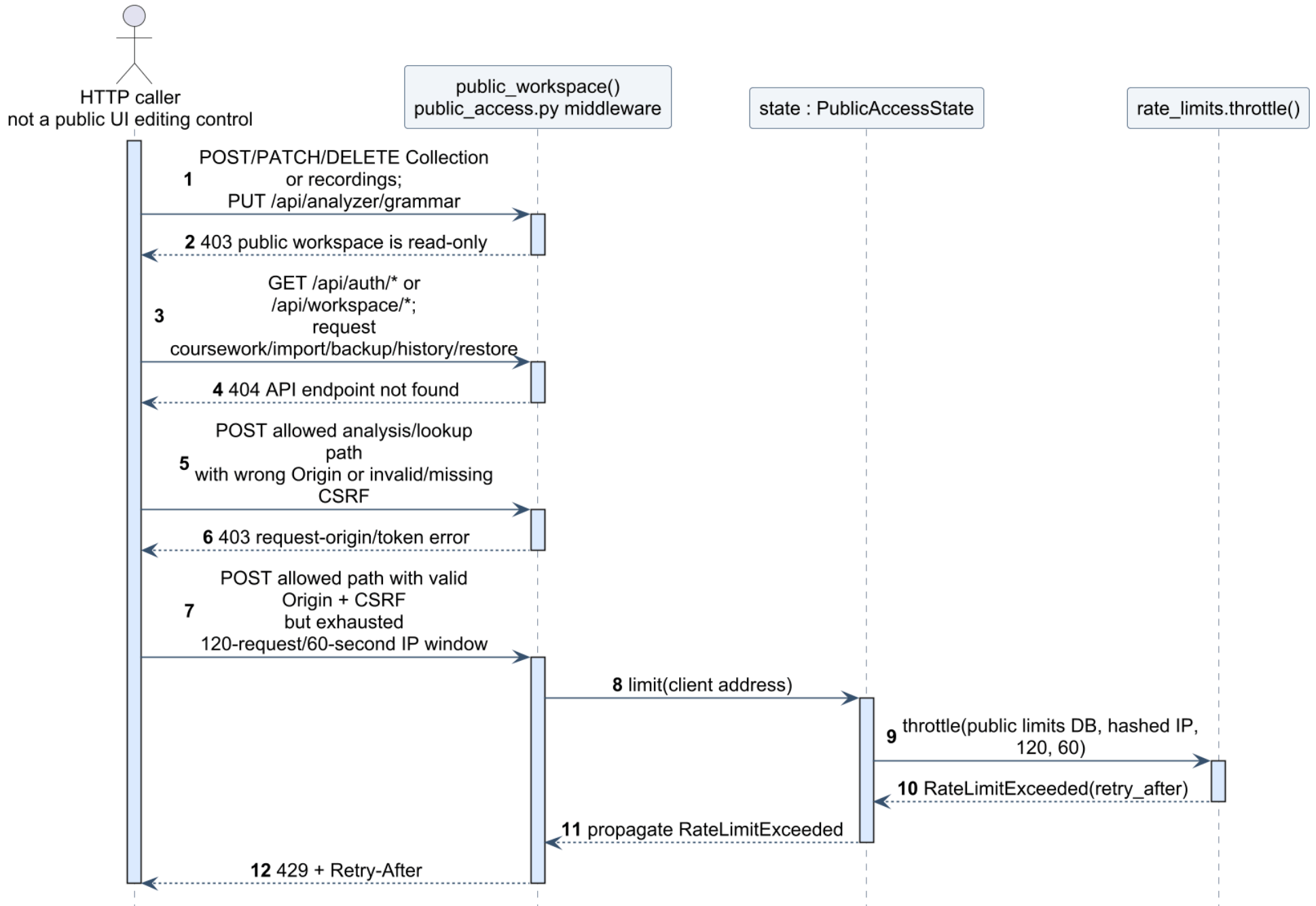
Dictionary reference data is not Collection and is not evidence of human collection.
The UI handoff sends no POST and never inserts a Collection entry.
The 938-entry merged dictionary is a dated source snapshot, not a test count or fieldwork result.

Existing recorded reading — lookup, guarded audio and honest absence



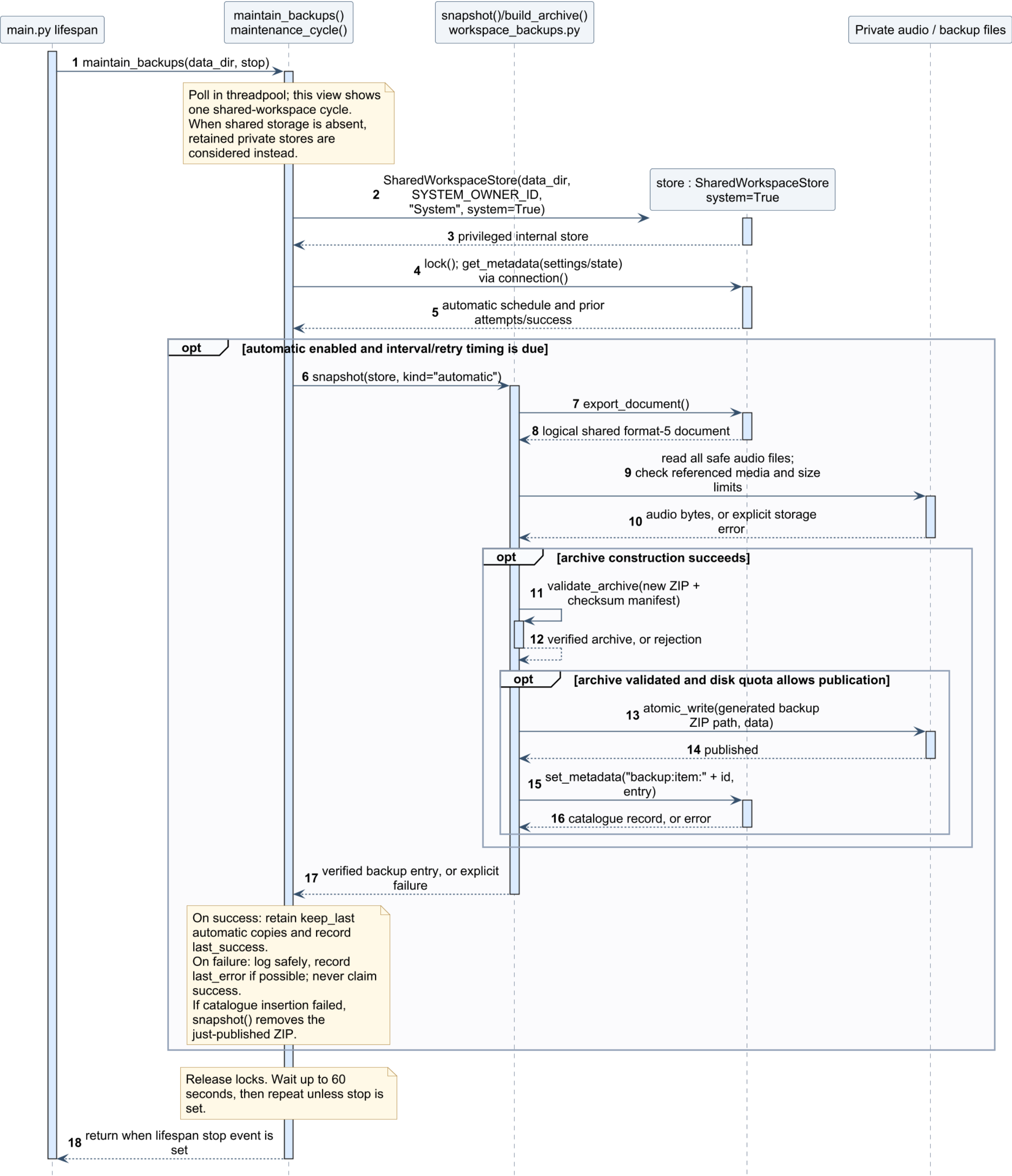
The browser hook name speak() selects a saved-reading lookup; it does not synthesize speech. Public ownership redaction never rewrites historical creator records. Collection attachment playback uses the separate read-only GET /api/dataset/{id}/audio route.

Public policy enforcement — rejected writes and hidden legacy routes



Allowed POST paths: /api/analyze, /api/analyzer/analyze, /api/analyzer/tests, /api/readings/lookup. Read-only-resource path checks distinguish forbidden writes (403) from absent legacy endpoints (404). These failures dispatch no domain mutation. PublicWorkspaceStore also rejects forbidden writes/imports. This sequence depicts separate requests; every call/reply and activation is independently balanced.

Operator background backup — one due shared-workspace cycle



No public backup API or recovery screen is mounted; maintenance can run privately in public mode. Default automatic=false; configured intervals are 24/168 hours and keep_last is 1..10. This backup preserves workspace/media, not auth.sqlite3 or /etc/camfranglais.env. Archive validation and atomic file publication precede a successful catalogue entry.